
Confidence as Control: A Survey of Confidence Utilization in Large Language Models

Yubo Li

Carnegie Mellon University

yubol@andrew.cmu.edu

Tianyang Zhou

Carnegie Mellon University

tianyanz@andrew.cmu.edu

Xiaobin Shen

Carnegie Mellon University

xiaobins@andrew.cmu.edu

Yidi Miao

Carnegie Mellon University

yidim@andrew.cmu.edu

Rema Padman

Carnegie Mellon University

rpadman@andrew.cmu.edu

Ramayya Krishnan

Carnegie Mellon University

rk2x@andrew.cmu.edu

Abstract

Most work on confidence in large language models has focused on estimation, uncertainty quantification, and calibration. In deployed systems, however, the key question is how confidence should be used to govern behavior. This survey studies **confidence utilization**: the use of confidence-related signals to control system decisions. We formalize this perspective through a unified framework in which confidence is defined over decision units under a local state and then consumed by a policy to determine actions. Using this lens, we organize the literature across full LLM lifecycle: training, inference, model selection and cascading, retrieval-augmented generation, risk management, and agentic control. We compare methods by signal source, decision unit, and functional role, and conclude by highlighting open challenges in confidence semantics, composition, source attribution, decision-aware evaluation, and robustness. Overall, the survey positions confidence not only as an estimation target, but as a control primitive for building more reliable and trustworthy LLM systems.

1 Introduction

Confidence in Large Language Models (LLMs) bridges the gap between internal uncertainty and actionable system behavior. A model that knows what it doesn't know is useful; a system that acts on this knowledge—retrieving when uncertain, deferring when unreliable, focusing learning where needed—is transformative. This survey concerns the latter. In this survey, we use *confidence* broadly to denote signals about the expected reliability or usefulness of a model decision, including uncertainty estimates, log-probability-based scores, verbalized confidence, sample agreement, semantic uncertainty, and verifier or reward-model scores. Recent work has shown that LLMs can expose useful self-knowledge through formulations such as $P(\text{True})$ and $P(\text{IK})$ Kadavath et al. (2022); that RLHF-tuned models can often express better-calibrated verbalized confidence when prompted appropriately Tian et al. (2023); that semantic entropy over meaning clusters helps detect confabulations in free-form generation Farquhar et al. (2024); and that agreement across sampled outputs can serve as a practical reliability proxy in reasoning and black-box factuality checking Wang et al. (2023a); Manakul et al. (2023). These advances have made confidence estimation increasingly usable in practice.

At the same time, several surveys and benchmarking studies have examined confidence and uncertainty in LLMs, predominantly focusing on estimation and calibration. Geng et al. (2024) provide a foundational taxonomy of confidence estimation and calibration techniques, while Shorinwa et al. (2025) and Liu et al. (2025c) offer comprehensive coverage of uncertainty quantification methods and their applications. Xie et al. (2024) focuses specifically on the calibration process for black-box LLMs, and Xiong et al. (2024) provide a systematic empirical evaluation of black-box confidence elicitation strategies. Taken together, this literature largely asks: how can we estimate, elicit, or calibrate high-quality confidence signals? Yet a good confidence score is not the end goal, it is the prerequisite. A systematic treatment of how confidence should actually govern system behavior remains absent.

Our survey addresses the critical next step: **how should systems utilize confidence?** As illustrated in Figure 1, we propose a taxonomy where confidence functions as *control*—not a passive measure of uncertainty, but an active governor that determines what to learn, how to reason, and when to defer. We trace this utilization across the full LLM lifecycle including: (i) training-time applications in data curation and alignment (§3), (ii) inference-time control over reasoning and decoding (§4), (iii) deployment-time decisions including model selection (§5), retrieval-augmented generation (RAG) (§6), and risk management (§7), and (iv) *agentic* settings (§8), where confidence becomes an actual control signal in multi-step loops: gating tool use and retries, triggering iterative refinement and backtracking, pruning search trees via verifier/process-reward confidence, and weighting votes in multi-agent debate. By synthesizing these distinct domains, we aim to offer a unified perspective for designing systems that act on uncertainty rather than merely quantify it.

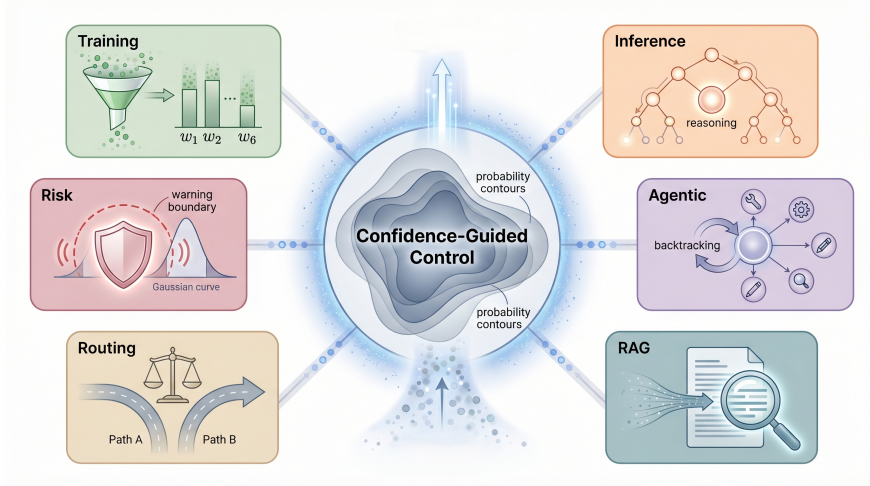


Figure 1: Confidence-guided control across six parallel domains of the LLM lifecycle.

2 Unified Definition and Notation

To compare methods across training, inference, routing, retrieval, risk control, and agentic systems, we adopt a single abstraction in which confidence is treated as a control signal. Throughout this survey, *confidence* is used in a deliberately broad sense: it may be a probability-like estimate, an uncertainty score, a log-probability-derived quantity, a verbalized confidence value, a sample-agreement statistic, a semantic uncertainty measure, a verifier or judge score, a reward-model score, or a hybrid of several such signals. The common feature is not a shared probabilistic semantics, but an operational one: the signal must affect a downstream decision.

Formally, let ξ_t denote the *decision state* at stage t . This state represents the full local context in which the system acts, and may include the input query, a partial generation, sampled candidates or reasoning traces, retrieved evidence, available tools or models, environment observations, memory, or a remaining compute budget. Let $U_t = \{u_1, \dots, u_n\}$ denote the set of *decision units* currently under consideration. A unit u may be a token, span, claim, retrieved chunk, training example, candidate response, model, tool, reasoning step,

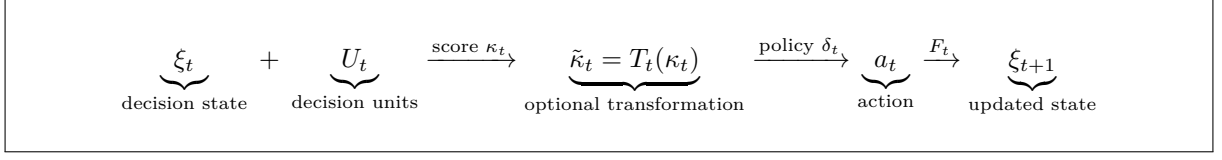


Figure 2: Unified confidence-as-control template. A confidence signal is defined over decision units under a local decision state, optionally transformed, and then consumed by a policy that determines the next action.

trajectory, or agent vote. This notation is intentionally more general than the conventional pair (x, y) because many confidence-utilization methods do not operate solely on completed outputs.

A *confidence signal* is any reliability-relevant score

$$\kappa_t(u; \xi_t) \in \mathbb{R}^m$$

assigned to a unit u under state ξ_t . Most methods use a scalar signal, but allowing κ_t to be vector-valued is convenient for systems that combine several sub-signals before acting. We refer to κ_t uniformly as “confidence” even when its concrete interpretation is uncertainty, expected correctness, expected utility, estimated helpfulness, or verifier-derived quality.

Confidence *utilization* is then the induced decision process

$$a_t \sim \delta_t(\cdot \mid \xi_t, U_t, \kappa_t), \quad \xi_{t+1} = F_t(\xi_t, a_t), \quad (1)$$

where δ_t is the policy that consumes the signal and produces an action a_t , and F_t updates the state after the action is executed. This perspective makes explicit that the object of study in this survey is not confidence estimation in isolation, but the way confidence participates in control.

In many systems, the raw score is not consumed directly. Instead it is transformed by calibration, normalization, aggregation, ranking, semantic clustering, or thresholding before the final decision is made. We therefore write

$$\tilde{\kappa}_t = T_t(\kappa_t), \quad a_t \sim \delta_t(\cdot \mid \xi_t, U_t, \tilde{\kappa}_t), \quad (2)$$

where T_t denotes an optional preprocessing map. Thresholding is thus only one special case of a broader pattern: confidence may be used to select, weight, allocate, aggregate, abstain, escalate, backtrack, or otherwise alter the control flow of the system.

This notation lets us characterize each method along three orthogonal axes. The first axis is the *source* of the signal: confidence may arise from the model itself through token probabilities, hidden states, or verbalized self-assessment; from sample-based behavior such as disagreement, self-consistency, or semantic entropy; from an auxiliary scorer such as a verifier, reward model, router, or judge; from external evidence or environment feedback such as retrieval signals and tool outcomes; or from an explicit hybridization of several sources. The second axis is the *unit or granularity* over which the signal is defined: token level, local content such as claims or chunks, item or candidate level, model/tool/agent level, step level, trajectory level, or full-episode level. The third axis is the *functional role* played by the signal in the downstream system: it may drive selection, weighting, resource allocation, control-flow decisions, aggregation, or serve directly as a learning signal.

With these axes in place, a method can be written compactly as

$$\kappa_t^{s,g}(u; \xi_t) \xrightarrow{T_t} \tilde{\kappa}_t \xrightarrow{\delta_t^r} a_t,$$

where s indexes the source, g indexes the unit or granularity, and r indexes the functional role. When the context is clear, we omit these superscripts.

This abstraction specializes naturally across the lifecycle considered in this survey. In training, the decision unit is often a training example, preference pair, or token, and confidence governs filtering, reweighting, or reward assignment. In inference, it is typically a candidate response, partial trace, or reasoning step, and confidence governs selection, stopping, continuation, or revision. In routing and cascading, the unit is usually

a model or route and the action is to choose, escalate, or defer. In retrieval-augmented generation, the relevant units include retrieval actions, chunks, and context sets, while the action may be to trigger retrieval, rerank evidence, filter passages, or halt. In risk management, the unit is often an answer, claim, or prediction set and confidence determines abstention, deferment, diagnosis, or coverage control. In agentic systems, the units may include tool invocations, branches, trajectories, or agent votes, and confidence may trigger search expansion, pruning, backtracking, replanning, or aggregation. The purpose of this framework is not to erase these differences, but to express them in a common language that makes the design space explicit.

3 Confidence-Aware Training

Training is the earliest point in the LLM lifecycle where confidence changes what the model learns rather than what it outputs. In the notation of §2, the decision state ξ_t now contains the current parameters, the minibatch, and any auxiliary teacher, reward, or judge models, while the decision units $u \in U_t$ may be full training examples, preference pairs, individual tokens, or complete rollouts. The resulting action is therefore training-specific: retain or discard a unit, assign it a loss weight, choose which part of a teacher signal to imitate, scale an RL reward, or teach the model to abstain. A central complication is that the semantics of the score vary substantially across papers. Some methods rely on self-confidence or uncertainty from the policy itself; others act on teacher confidence, reward-model uncertainty, self-reflection scores, or broader difficulty and utility proxies. The common structure is still the same: a reliability-relevant score over training units determines how gradient mass is allocated. Table 1 condenses these training-stage methods by signal source, unit, and update role.

3.1 Confidence-aware data selection

Training-time confidence first appears as a curation signal. A large cluster of papers interprets low confidence, or a closely related difficulty score, as evidence of learning value. Li et al. (2024c) define instruction-following difficulty from the gap between answer likelihood with and without the instruction, and use that score to retain examples that are challenging in a model-specific way. Li et al. (2024b) show that this ranking transfers surprisingly well from weak to strong models, so a much smaller proxy model can perform the expensive filtering step. Han et al. (2025) combines uncertainty with graph-based influence, favoring examples that are both hard and structurally representative. In a nearby active-learning setting, Muldrew et al. (2024) use predictive entropy and preference certainty to decide which completion pairs deserve expensive preference labels. Across these methods, low confidence does not mean bad data; it often means that a sample lies near the model’s current frontier of learnability.

Other data-selection methods treat consistency or judged quality, rather than raw difficulty, as the more useful signal. Liu et al. (2024a) ranks instruction data using uncertainty-aware self-reflection across score tokens, paraphrased prompts, and multiple model scales, thereby rewarding examples whose quality assessment is both strong and stable. Chen & Mueller (2024) similarly uses a confidence-bearing evaluator to filter noisy supervision and to rewrite targets only when the correction itself is judged reliable. Sachdeva et al. (2024) moves the same logic to an external judge model, using the probability of a positive usefulness judgment as a quality score. These papers are closer to confidence-guided quality assessment than to pure difficulty filtering, because the signal is interpreted as expected usefulness or trustworthiness rather than mere challenge.

The neighboring curation literature is important context, but it should not be conflated with confidence-aware training. Zhou et al. (2023) is a quality-first human curation baseline rather than a confidence method; Abbas et al. (2023) prunes redundancy through semantic similarity rather than uncertainty; Zhang et al. (2025b) scores examples by estimated holdout-loss impact; and Pan et al. (2025) uses attribution to identify unsafe training data. Pang et al. (2025) further shifts the decision unit below the sample level by removing low-value tokens inside otherwise useful examples, but its signal is closer to token influence than to confidence. We retain these works as nearby baselines because they solve the same control problem—which data should shape the model—even when the scoring semantics are not confidence in the strict sense.

3.2 Confidence-aware fine-tuning, distillation, and abstention tuning

Once the data has been chosen, confidence can act directly on the update rule. Krishnan et al. (2024) provide the clearest policy-side example: they augment causal language modeling with an uncertainty-aware objective that encourages low uncertainty on correct tokens and high uncertainty on incorrect tokens, with the explicit aim of making downstream uncertainty signals more usable for hallucination detection and selective generation. Li et al. (2025c) uses contextual uncertainty differently. Its two-stage procedure first teaches the model to recognize when the provided evidence is insufficient, and then separately teaches compliance with abstention instructions, so that uncertainty becomes an answer-versus-refuse decision rather than only a calibration statistic. By contrast, Rahmati et al. (2025) mainly improve uncertainty estimation under parameter-efficient fine-tuning through contextual stochastic adapters; it is therefore best viewed as an estimation-improving companion to this section rather than a core utilization method.

Teacher-student settings make the control role even more explicit. Huang et al. (2025b) distill only where the teacher’s propose-and-verify signal is trustworthy, so teacher confidence determines which tokens receive supervision. Zhong et al. (2024) likewise uses teacher token uncertainty to split easy and hard tokens and to vary the distillation mode accordingly. In both cases, the confidence source is external to the student policy: the score decides where imitation should be sharp, softened, or withheld. Other training-control papers occupy the boundary of the category. Li et al. (2024a) uses student-specific difficulty and compatibility statistics to select teacher-refined data, and Pang et al. (2025) uses token-level influence to prune within-example supervision. These methods reinforce the broader point that training-time control often lives at finer granularities than whole samples, even when the operative signal is only confidence-adjacent.

3.3 Confidence-aware preference optimization and RL

Confidence becomes most explicit in preference optimization and RL, where it can enter as a reward, a penalty, or a gating signal over updates. Several DPO-family baselines already optimize probability-derived surrogates: DPO uses policy-relative log-probability ratios Rafailov et al. (2023), SimPO replaces the reference-based reward with average log probability Meng et al. (2024), and β -DPO adapts optimization strength across batches Wu et al. (2024). These methods are important background because they show that likelihood-derived quantities already function as training signals. However, later work makes the confidence role explicit rather than implicit.

On the policy side, Yoon et al. (2025) select low-confidence tokens as the sites where preference optimization should act, based on the observation that these tokens carry larger alignment gradients. Pokharel et al. (2025) modulate multilingual preference updates with a relative reward margin, treating stronger pairwise preference gaps as more trustworthy supervision. Lu et al. (2025) uses local low-confidence points inside a reasoning trace to decide where to split, branch, and construct preference pairs. Du et al. (2025) turns final-answer confidence into a reward proxy for close-ended reasoning and then reuses confidence gaps to build stronger DPO data. In sequence-level RL, Li et al. (2025d) sharpen the policy around its own high-probability responses, Prabhudesai et al. (2025) use negative entropy as an intrinsic reward, Zhou et al. (2025a) reweight trajectories by sequence confidence and problem difficulty, and Liu et al. (2025b) jointly optimize reasoning accuracy and sequence-level calibration so that confidence remains useful after post-training. Taken together, these papers show that confidence can decide not only which output is preferred, but also where along a response or trajectory learning pressure should be concentrated.

A second line of work places the uncertainty in the supervision signal rather than in the policy output itself. Zhai et al. (2024) penalize rewards with high ensemble uncertainty so the policy does not overoptimize dubious reward spikes, while Banerjee & Gopalan (2024) derive a variance-aware conservative policy objective from the same intuition. Leng et al. (2024) locate the problem in the reward pipeline itself, showing that RLHF reward models prefer overly confident responses and then correcting this bias either in reward-model training or in reward calculation during PPO. Wu et al. (2025b) goes one step further and trains the model to answer only when its estimated correctness exceeds a user-specified risk tolerance, thereby making confidence operationally identical to an abstention policy. Not every token-level alignment refinement is confidence-aware, however: Liu et al. (2025a), for example, reweight tokens by estimated importance rather than reliability.

Method	Source	Signal	Unit	Training Action
Data curation / fine-tuning / distillation				
Cherry (Li et al., 2024c)	self	IFD / instruction-conditioned likelihood gap	example	select SFT examples
Superfiltering (Li et al., 2024b)	self	weak-model IFD / difficulty ranking	example	proxy-filter examples for strong model
UniMax (Han et al., 2025)	hybrid	uncertainty plus graph influence	example	select representative hard examples
Active-Pref (Muldwrew et al., 2024)	hybrid	predictive entropy plus preference certainty	pair	acquire preference labels
SelectIT (Liu et al., 2024a)	hybrid	self-reflection consistency across tokens / prompts / models	example	select instruction data
CLEAR (Chen & Mueller, 2024)	hybrid	BSDetector confidence from agreement and self-reflection	example	filter and rectify supervision
Ask-LLM (Sachdeva et al., 2024)	auxiliary	judge usefulness probability $P(\text{YES})$	example	select pretraining data
UA-CLM (Krishnan et al., 2024)	self	token entropy / white-box uncertainty	token	reweight CLM loss for calibration
US-Tuning (Li et al., 2025c)	self	context-sufficiency uncertainty	question-context pair	tune answer-versus-abstain behavior
C-LoRA (Rahmati et al., 2025)	self	contextual posterior predictive uncertainty	example	improve PEFT uncertainty
SelecTKD (Huang et al., 2025b)	auxiliary	teacher propose-verify acceptance confidence	token	distill only trusted tokens
ATKD (Zhong et al., 2024)	auxiliary	teacher token uncertainty / difficulty coefficient	token	switch distillation mode by token
SRD (Li et al., 2024a)	self	IFD plus r-IFD difficulty / compatibility	example	select teacher-refined data
Preference optimization / RL				
DPO (Rafailov et al., 2023)	self	policy-relative log-probability ratio	pair	preference optimization baseline
SimPO (Meng et al., 2024)	self	average log probability	pair	reference-free preference optimization baseline
β -DPO (Wu et al., 2024)	self	batch informativeness / reward discrepancy	batch / pair	adapt optimization strength
ConfPO (Yoon et al., 2025)	self	below-average token probability	token	select alignment-critical tokens
CAPO (Pokharel et al., 2025)	hybrid	relative reward margin as pair confidence	pair	scale preference loss by strength
CGPO (Lu et al., 2025)	hybrid	token confidence + RM score	step	branch traces and build pairs
CRew (Du et al., 2025)	self	final-answer token confidence	answer / response	reward proxy and pair construction
RLSC (Li et al., 2025d)	self	old-policy response probability	response	self-confidence reward shaping
RENT (Prabhudesai et al., 2025)	self	negative entropy on late-response tokens	late tokens / trajectory	intrinsic RL reward
CoDaPO (Zhou et al., 2025a)	hybrid	sequence confidence plus problem difficulty	trajectory	reweight RL advantages
C2GSPG (Liu et al., 2025b)	self	normalized sequence probability	trajectory	calibration-aware sequence RL
UP-RLHF (Zhai et al., 2024)	auxiliary	reward-model ensemble variance	response	penalize uncertain rewards
UA-RLHF (Banerjee & Gopalan, 2024)	auxiliary	reward-model ensemble variance	response	conservative variance-aware RLHF
Taming-OC (Leng et al., 2024)	hybrid	verbalized confidence in reward calibration	response	calibrate reward model / reward score
BCHRL (Wu et al., 2025b)	self	verbalized confidence or critic value	response / claim	train risk-sensitive abstention policy

Table 1: Training-stage confidence-utilization methods, organized by signal source, signal form, decision unit, and training action. The table merges data curation, fine-tuning, distillation, preference optimization, and RL into one grouped comparison.

This distinction is useful because it separates confidence-guided control from the broader family of fine-grained optimization heuristics.

Discussion. Across these families, training uses confidence to allocate gradient mass. The same arithmetic operation can therefore mean very different things depending on the source of the signal. Low self-confidence may mark high-learning-value examples or tokens, as in Cherry, ConfPO, or CGPO. High external confidence may mark trustworthy supervision, as in SelecTKD, CLEAR, or ASK-LLM. High reward uncertainty, by contrast, often suppresses optimization, as in UP-RLHF and UA-RLHF. This is precisely why the source/unit/role decomposition from §2 is most useful in the training setting: without it, difficulty, quality, preference strength, teacher reliability, and reward uncertainty all collapse into a single overloaded notion of confidence. The main lesson of this section is therefore structural rather than metric-specific. During training, confidence matters because it determines where learning should occur, where it should be damped, and when the model should be explicitly taught to abstain; in later sections, the same family of signals will shift from controlling gradient allocation to controlling system behavior at deployment time.

4 Confidence-Driven Inference

Inference is where confidence becomes an online control variable. In the notation of §2, the decision state ξ_t now contains the prompt, partial generations, sampled candidates, any verifier or judge outputs, and the remaining inference budget, while the decision units $u \in U_t$ may be complete candidate responses, partial reasoning states, token groups, or next-token alternatives. The resulting action therefore depends on where the signal is injected: at the candidate level it selects or aggregates complete outputs, at the state level it decides whether to continue, stop, refine, or backtrack, and at the token level it reshapes the predictive distribution itself. The literature in this section spans both explicit confidence signals, such as answer agreement or elicited confidence, and broader reliability surrogates, such as verifier scores, local log-probability drops, or disagreement between predictive views. We organize the section accordingly. Table 2 provides a compact view of the main inference-stage methods and the control decisions they support.

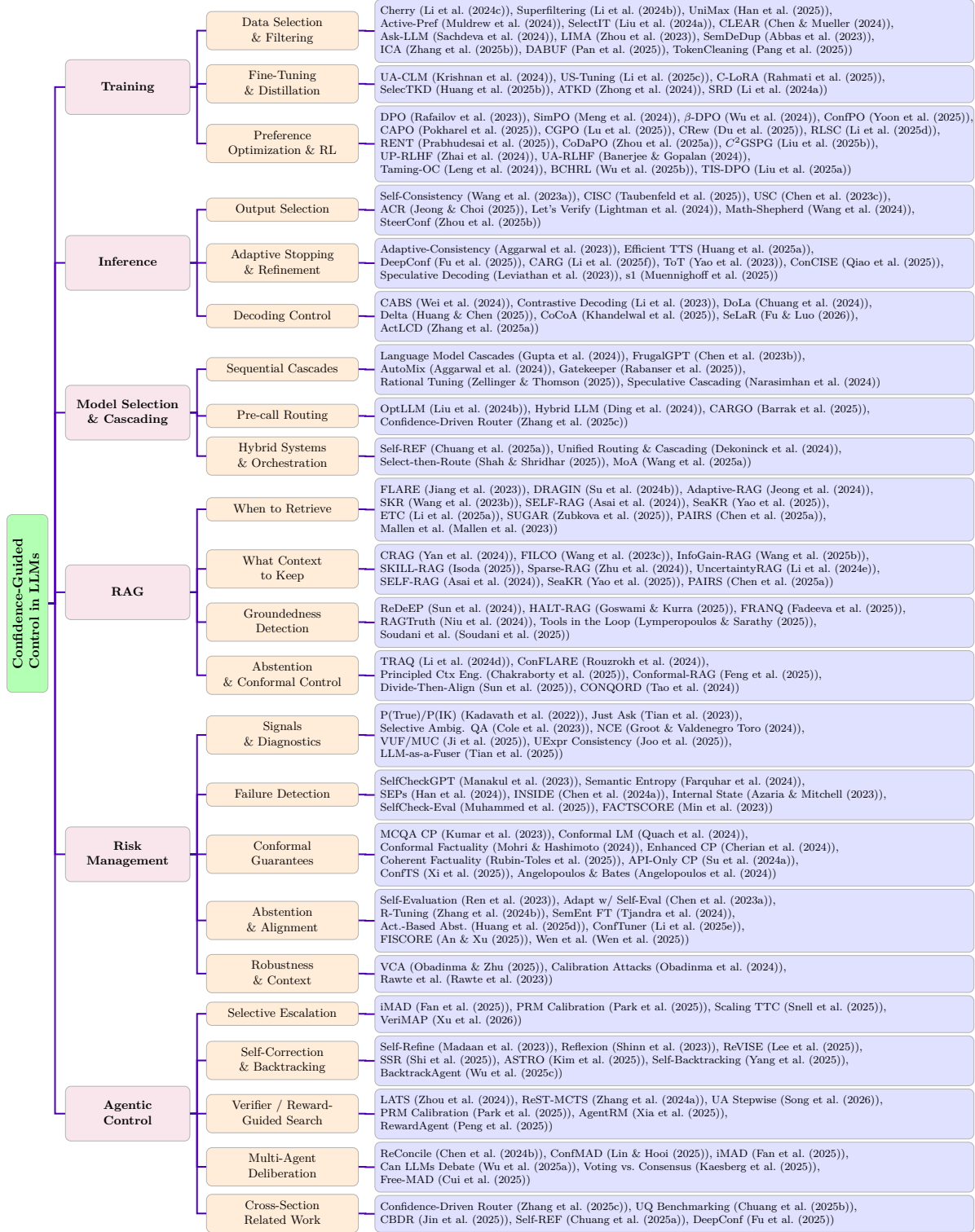


Figure 3: Taxonomy of confidence utilization in LLMs across six domains, with leaves listing core and adjacent cited methods discussed in the survey.

4.1 Confidence-driven output selection

The most direct use of inference-time confidence acts on a fixed candidate set. Wang et al. (2023a) establish the basic template with self-consistency: sample diverse reasoning paths, extract their final answers, and use answer agreement as an implicit confidence signal for final selection. Later work asks how to make this agreement signal more informative. Taubenfeld et al. (2025) augment self-consistency with path-level confidence scores and show that the decisive property is not global calibration, but *within-question discrimination*: confidence must separate stronger and weaker responses to the same question if it is to improve multi-sample selection. Their CISC procedure therefore normalizes candidate confidence within each question and performs confidence-weighted voting, with $P(\text{True})$ emerging as especially effective for this use case. In the same spirit, Chen et al. (2023c) remove the requirement that answers be cleanly extractable. Universal Self-Consistency asks the model itself to choose the most consistent response among sampled candidates, so the confidence signal is comparative rather than scalar: the model acts as a consistency-based selector over an answer set rather than producing an independent score for each answer.

This family also supports more selective use of confidence. Jeong & Choi (2025) begin from an ordinary self-consistency distribution and trigger an additional re-scoring stage only when that distribution is too flat to be trusted. The first-stage candidate frequencies define an ambiguity signal; only ambiguous cases are passed to a second-stage multiple-choice selection step, whose outputs are then combined with the original self-consistency scores. Confidence here is therefore used twice: first to detect when majority voting is unreliable, and then to support final answer choice among the surviving candidates.

An orthogonal line of work replaces self-derived confidence with externally trained verifier signals. Lightman et al. (2024) show that process reward models trained with step-level supervision are stronger search-time selectors than outcome-only reward models, precisely because they can discriminate errors that occur inside otherwise plausible full solutions. Wang et al. (2024) reduce the annotation burden by constructing process labels automatically, but the inference role is similar: a step-level verifier scores candidate solutions and the aggregate verifier score determines which completed solution should survive. These papers belong in the inference section because the verifier signal governs selection at test time, even though the signal itself is produced by a separately trained auxiliary model.

Finally, some methods are better understood as confidence-elicitation primitives that support downstream selection. Zhou et al. (2025b) use prompt steering to obtain multiple verbalized confidence views of the same answer, combine mean confidence with answer consistency and confidence consistency, and then use the resulting calibrated score for answer selection. This makes SteerConf highly relevant to inference, but its main contribution is black-box confidence elicitation rather than multi-candidate reasoning control in the narrower self-consistency sense.

4.2 Adaptive stopping, refinement, and search

Confidence can also act before the candidate set is complete. In this regime, the relevant units are partial answer distributions, intermediate reasoning segments, or search states, and the action is to continue, stop, refine, or revisit the current trajectory. Aggarwal et al. (2023) provide the cleanest agreement-based example: Adaptive-Consistency samples one candidate at a time and stops when the estimated probability that the current majority answer will remain dominant is sufficiently high. Confidence is therefore not used to rank completed outputs, but to determine whether additional sampling is still worth the compute. Huang et al. (2025a) push the same logic further by first distilling self-consistency-derived confidence into the model itself and then using the calibrated score for weighted voting, early stopping in Best-of- N , and adaptive self-consistency. The paper thus straddles post-training and inference, but its operational contribution is squarely inference-time budget control.

Fu et al. (2025) show that the relevant signal need not live at the full-trace level. Their DeepConf framework derives confidence from internal token probabilities and finds that local low-confidence regions, especially the least-confident token groups, are more diagnostic than whole-trace averages. This signal then supports two distinct actions: offline confidence-weighted filtering or voting over completed traces, and online early stopping when a running trace enters a sufficiently weak local segment. Li et al. (2025f) study a different inference

setting—sequential interactions rather than one-shot reasoning—but the control logic is similar. Their CARG framework extracts response confidence from answer-token probabilities and uses it to decide whether a model should maintain its current answer or reconsider it under later follow-up pressure. Confidence here functions as a persistence-versus-revision signal over turns.

More broadly, search-style reasoning methods also employ confidence-like state evaluation. Yao et al. (2023) use self-value and self-vote prompts to score partial thought states, then use those scores to expand promising branches and prune weak ones. Under the broad operational definition adopted in this survey, these state-evaluation scores function as confidence over incomplete reasoning states rather than as confidence over completed answers. Qiao et al. (2025) is a boundary case between inference and training: it uses confidence injection and stopping heuristics to suppress redundant reflection and construct concise reasoning traces, but much of the eventual gain comes from fine-tuning on the compressed data rather than from leaving the control rule purely at inference time.

This subsection also clarifies an important boundary of scope. Nearby test-time scaling papers such as speculative decoding Leviathan et al. (2023) and budget forcing in *s1* Muennighoff et al. (2025) adapt compute at inference time, but they do not do so on the basis of a reliability signal in the sense of this survey. We therefore treat them as adjacent compute-control baselines rather than as core confidence-utilization methods.

4.3 Confidence-shaped decoding control

At the finest scale, the decision unit is the next-token distribution itself. Here many methods do not estimate an explicit scalar confidence over a completed answer; instead they use disagreement between predictive views as an implicit token-level surrogate for reliability. Li et al. (2023) provide the canonical example. Contrastive decoding compares an expert model with a weaker amateur model and prefers tokens that the expert favors but the amateur over-amplifies less strongly, subject to a plausibility constraint that prevents the contrastive objective from selecting implausible tokens. The signal is therefore an expert–amateur disagreement score that directly reweights token choices. Chuang et al. (2024) internalize this contrast within a single network by comparing late and early layers, using layer disagreement as a token-level control signal for factual decoding. Zhang et al. (2025a) make this control adaptive: rather than applying layer contrast at every token, ActLCD learns a policy for when contrastive intervention is actually needed.

Other methods make the underlying confidence interpretation more explicit. Huang & Chen (2025) use masked-context contrast to identify tokens whose support is fragile under perturbations of the provided context, so the signal should be understood as contextual fragility rather than general answer correctness. Khandelwal et al. (2025) similarly frame token control as adaptive trust in context: they combine prior–context divergence, entropy gap, and contextual peakedness into a gating rule that determines how strongly decoding should follow the context-conditioned distribution rather than the model’s prior distribution. The resulting signal is hybrid by construction, since it is defined by the relation between parametric and context-conditioned views rather than by either one alone. Fu & Luo (2026) extend this token-level control view to latent reasoning: normalized top- k entropy gates whether decoding should preserve the ordinary discrete token embedding at high-confidence steps or activate soft latent embeddings at low-confidence exploratory steps, with an entropy-aware contrastive regularizer used to keep multiple latent alternatives alive. Finally, Wei et al. (2024) show that the relevant inference unit need not be a token at all. Their CABS framework learns a hidden-state-based confidence model over generated sub-structures and then uses those local confidence scores to guide beam search in structured generation, underscoring that the unit axis from §2 remains important even inside a single inference stage.

Discussion. Across these families, inference shifts confidence from gradient allocation to online control. Candidate-level methods primarily aggregate or select among completed outputs; state-level methods determine whether reasoning should continue, stop, branch, or be revised; token-level methods reshape the predictive distribution directly. The same caution from §3 reappears, however: confidence is not semantically uniform across these papers. Agreement frequency, $P(\text{True})$, verbalized confidence, verifier scores, local log-probability drops, and contrastive disagreement are not interchangeable, even if they all influence the next inference action. Making these distinctions explicit is what separates core confidence-driven reasoning methods from adjacent compute-control or decoding heuristics. It also prepares the transition to routing and cascading,

Method	Source	Signal	Unit	Role	Access
Output selection					
Self-Consistency (Wang et al., 2023a)	self	answer agreement	candidate answer	vote	MS
CISC (Taubenfeld et al., 2025)	self	$P(\text{True})$ / path confidence	path / candidate	weighted vote	MS
Universal SC (Chen et al., 2023c)	self	comparative consistency judgment	candidate response	select	MS
ACR (Jeong & Choi, 2025)	self	SC frequency + ambiguity score	candidate answer	trigger, rescore	MS
PRM (Lightman et al., 2024)	auxiliary	process reward score	step / solution	rerank	AV, MS
Math-Shepherd (Wang et al., 2024)	auxiliary	PRM step score	step / solution	rerank	AV, MS
SteerConf (Zhou et al., 2025b)	self	steered verbal confidence	answer	calibrate, select	BB
Adaptive stopping, refinement, and search					
Adaptive-Consistency (Aggarwal et al., 2023)	self	majority stability	answer set	stop	MS
Efficient TTS (Huang et al., 2025a)	self	calibrated response confidence	candidate answer	vote, stop	MS, FT
DeepConf (Fu et al., 2025)	self	lowest-group logprob	token group / trace	filter, stop	WB, MS
Firm-or-Fickle (Li et al., 2025f)	self	answer-token probability	response / turn	maintain, revise	WB
ToT (Yao et al., 2023)	self	self-value / self-vote	thought state	expand, prune	MS
ConCISE (Qiao et al., 2025)	self	reflection confidence	step / trajectory	stop, compress	FT
Confidence-shaped decoding control					
Contrastive Decoding (Li et al., 2023)	hybrid	expert–amateur gap	token	reweight	2M
DoLa (Chuang et al., 2024)	self	layer contrast	token	reweight	WB
Delta-CD (Huang & Chen, 2025)	self	masked-context contrast	token	reweight	WB
CoCoA (Khandelwal et al., 2025)	hybrid	prior-context conflict	token	blend, reweight	WB
SeLaR (Fu & Luo, 2026)	self	top- k entropy	token / latent step	gate, regularize	WB
ActLCD (Zhang et al., 2025a)	self	learned contrast trigger	token	gate, reweight	WB, FT
CABS (Wei et al., 2024)	self	sub-structure confidence	sub-structure	beam rerank	WB, FT

Table 2: Inference-stage confidence-utilization methods, organized by signal source, signal form, decision unit, functional role, and operational access.

Access: MS = multiple samples or search branches; WB = white-box logits or hidden states; AV = auxiliary verifier; BB = black-box prompting; 2M = second model at inference; FT = extra post-training or learned controller. Adjacent compute-control baselines such as speculative decoding and *sl* are discussed in the text but omitted here because they do not use a reliability signal in the sense of this survey.

where the decision unit changes again—from candidate outputs within one model to model choices across a portfolio.

5 Confidence-Guided Model Selection

This section studies confidence-guided model selection in systems with multiple candidate models. Under the notation of Section 2, the decision state includes the query, the available model pool, the remaining budget or latency allowance, and, in some methods, a provisional answer or partial generation. The decision units are therefore models, cascade stages, or token-level hand-off points, depending on when the system intervenes. The confidence signal may come from the small model itself, from an external router, from judge or verifier signals, or from uncertainty summaries such as semantic entropy. The downstream action is to select a model, defer to a stronger one, continue with the current model, or aggregate multiple outputs.

Relative to earlier sections, model selection introduces a particularly important timing distinction. Some methods make an ex-ante decision before any candidate model is run; others make a post-hoc deferral decision after a cheaper model has already produced an answer; and a smaller set intervenes mid-generation by handing off or verifying partial outputs. This timing axis determines both which confidence signals are available and what the signal means. Pre-call routing typically uses predicted suitability or expected quality, whereas post-call deferral is able to condition on an actual candidate answer and therefore use answer-conditioned uncertainty or verification scores. Table 3 organizes these routing and cascade methods by timing, signal source, and downstream action.

5.1 Confidence-guided selection architectures

Deploying heterogeneous LLMs creates a persistent cost–quality tradeoff: stronger models are usually more reliable, but they are also slower or more expensive. Confidence-guided selection addresses this tradeoff by making the amount of computation conditional on the query and, in some cases, on a provisional answer. Earlier NLP work on model cascading already showed that instance-dependent escalation across models can jointly improve efficiency and accuracy (Varshney & Baral, 2022); the recent LLM literature inherits this

same control problem, but with richer confidence signals and sharper cost-quality tradeoffs. The literature in this section falls into three broad families: sequential cascades that defer uncertain cases to stronger models, pre-call routers that predict which model is most suitable before generation, and hybrid systems that combine both mechanisms.

Sequential cascading: deferral as answer-conditioned escalation. Sequential cascades run cheaper models first and defer only uncertain cases to stronger models. A core question in this setting is how to summarize the weaker model’s uncertainty in a way that is useful for the deferral decision. Gupta et al. (2024) show that naive sequence-level confidence summaries are strongly length-biased in generative settings and argue that token-level uncertainty should be treated as the primitive signal. Their main contribution is to use quantiles over token-level uncertainty, together with learned post-hoc deferral rules, rather than collapsing a response too early into a single length-sensitive score.

Chen et al. (2023b) broaden the picture from one cascade rule to a larger cost-aware orchestration framework. FrugalGPT studies prompt adaptation, model approximation, and cascades under a unified cost-quality objective, with adaptive model selection as its central systems contribution. In the present survey, it is best viewed as a foundational routing-and-cascade system: it demonstrates that adaptive orchestration across heterogeneous APIs can move the cost-quality frontier substantially, even when the control signal is a learned utility estimate rather than an explicitly calibrated confidence score.

Aggarwal et al. (2024) are closer to a direct confidence-utilization view. AutoMix generates an initial answer with a cheaper model, then asks that model to self-verify its own output using an entailment-style verification prompt. Because this self-verification signal is noisy, the routing policy is formulated as a POMDP over latent query difficulty rather than as a simple fixed-threshold rule. The key lesson is that even imperfect confidence observations can drive effective deferral when the downstream controller explicitly models uncertainty about the signal itself.

Two later papers refine cascade control in complementary ways. Rabanser et al. (2025) improve cascades upstream by fine-tuning the smaller model so that its confidence is more useful for deferral, emphasizing that cascade quality can be improved by confidence tuning rather than threshold tuning alone. Zellinger & Thomson (2025) focus instead on the downstream decision rule: they calibrate raw model confidences and model their dependence across cascade stages, then optimize thresholds under an explicit error-cost objective. Together, these papers show that better cascading can come either from better confidence signals or from more principled use of those signals.

Narasimhan et al. (2024) connect cascading to speculative decoding. Their speculative cascade view treats the draft model and the verifier model as a joint system in which standard cascade deferral rules can be implemented through speculative execution. This makes the paper relevant to both model selection and inference efficiency: the confidence signal still governs whether the stronger model must intervene, but the intervention is implemented at the token level rather than only after a full answer is produced.

Pre-call routing: confidence as model suitability. In pre-call routing, the system chooses a model before any candidate answer is observed. The confidence signal is therefore not answer correctness in the usual sense, but an estimate of expected model suitability for the query under a target budget or quality level. Liu et al. (2024b) make this perspective explicit by casting routing as multi-objective assignment under uncertainty. OptLLM combines a predictive model over candidate LLM suitability with bootstrap-based uncertainty estimates and then searches for Pareto-efficient assignments under cost and performance constraints. It is thus less a calibration paper than an uncertainty-aware optimization layer over model assignment.

RouteLLM makes this logic especially concrete by training routers from human preference data to choose between stronger and weaker models, showing that relative preference supervision can produce cost-sensitive pre-call routing policies that transfer across changed model pairs (Ong et al., 2024).

Ding et al. (2024) study a more focused two-model setting. Their router predicts the quality gap between a small and a large model and conditions the routing decision on a desired response-quality target. This is an important distinction from generic difficulty estimation: the operative signal is not simply whether a query is hard, but whether the smaller model is expected to be close enough to the larger one for the user’s quality

requirement. The paper also models uncertainty arising from stochastic generation, which makes the routing rule more robust than a purely deterministic classifier.

Barrak et al. (2025) extend routing to a pool of expert models. CARGO trains an embedding-based regressor on pairwise LLM-judge preferences to predict each candidate model’s expected performance, then uses the gap between the top predicted scores as an operational confidence measure. When the gap is small, an auxiliary binary classifier is invoked to resolve the ambiguity. The confidence signal here is therefore relative and comparative: it measures how decisively the router prefers one model over another, rather than how likely any single answer is to be correct.

A different route to confidence-guided routing is developed by Zhang et al. (2025c), who use semantic entropy in an edge-cloud setting. The local model first produces multiple candidates, which are grouped by semantic equivalence; entropy over these meaning-level clusters is then used as the uncertainty signal that triggers offloading to a stronger cloud model. This paper is noteworthy because it directly connects modern uncertainty estimation methods to routing decisions and emphasizes response quality, not only benchmark accuracy, as the quantity to preserve.

Chuang et al. (2025a) occupy a useful middle position between routing and confidence learning. Self-REF fine-tunes a local model to emit confidence tokens that are conditioned on the answer it has already produced, and the resulting token probabilities are then used for routing or rejection. Although the method is learned during training, its main downstream role is post-answer model selection: the system decides whether the local model should be trusted or whether a stronger fallback model should be invoked.

Hybrid systems and adjacent orchestration methods. Several recent systems explicitly combine ex-ante routing with post-hoc deferral. Dekoninck et al. (2024) formalize this relationship by distinguishing ex-ante quality estimation, which supports routing before a model is called, from post-hoc quality estimation, which supports cascading after an answer is observed. Their unified cascade-routing view is important for the survey because it shows that routing and cascading are not disjoint families but two points in a common decision space.

Shah & Shridhar (2025) provide a practical realization of this hybrid view. SELECT-THEN-ROUTE first narrows the candidate set through a taxonomy-guided router, then applies confidence-triggered escalation within the shortlisted pool using multi-judge agreement. The first stage reduces the decision space; the second stage uses a confidence signal to decide whether a cheaper selected model is sufficient. This decomposition is representative of more realistic production systems, where model selection is often hierarchical rather than a single one-shot choice.

Finally, some multi-model systems are adjacent rather than central to confidence-guided routing. Wang et al. (2025a), for example, study collaborative aggregation across models in a layered mixture-of-agents architecture. MoA is useful context for multi-model orchestration, but its main mechanism is iterative synthesis rather than explicit confidence-based model selection. We therefore treat it as a neighboring orchestration pattern rather than a core routing or cascade method.

Discussion. Confidence-guided model selection extends the same basic pattern seen in earlier sections: a reliability-relevant signal is estimated for a decision unit and then converted into an action. What changes in this section is the unit itself. Instead of deciding which token, step, or answer to keep, the system now decides which model to call, whether to defer to a stronger one, or whether to continue a multi-stage pipeline. This shift makes confidence less about a single model’s internal certainty and more about comparative suitability under resource constraints.

The routing literature is especially clarified by separating when the decision is made. Pre-call routers such as OptLLM, RouteLLM, Hybrid LLM, and CARGO operate without seeing an answer and therefore rely on predicted model quality, predictive uncertainty, or score margins. Post-hoc deferral methods such as Language Model Cascades, AutoMix, Self-REF, and semantic-entropy offloading observe a provisional answer first and can therefore use answer-conditioned signals. Mid-generation systems such as speculative cascading occupy an intermediate point in which the confidence signal is tied to partial generations and verifier intervention.

Method	Source	Signal	Unit	Role	Access
Language Model Cascades (Gupta et al., 2024)	self	token-uncertainty quantiles	answer	defer	Post
FrugalGPT (Chen et al., 2023b)	ext	predicted utility	query / answer	route, defer	Pre, Post, Aux
AutoMix (Aggarwal et al., 2024)	hybrid	self-verification entailment	answer	defer	Post, Aux
Gatekeeper (Rabanser et al., 2025)	self	tuned small-model confidence	answer	defer	Post, FT
Rational Tuning (Zellinger & Thomson, 2025)	self	calibrated confidence + copula	stage	stop	Post
OptLLM (Liu et al., 2024b)	ext	bootstrap suitability uncertainty	query	route	Pre, Aux
Hybrid LLM (Ding et al., 2024)	ext	predicted quality gap	query	route	Pre, Aux
CARGO (Barrak et al., 2025)	judge/ext	score gap + tie-break classifier	query	route	Pre, Aux
Self-REF (Chuang et al., 2025a)	self	confidence-token probability	answer	route, reject	Post, FT
Semantic Entropy (Zhang et al., 2025c)	self	semantic entropy	query / answer	offload	Post, MS
Unified Routing & Cascading (Dekoninck et al., 2024)	hybrid	ex-ante + post-hoc quality	query / answer	route, defer	Pre, Post
Select-then-Route (Shah & Shridhar, 2025)	hybrid	taxonomy + judge agreement	query / answer	shortlist, defer	Pre, Post, Aux
Speculative Cascading (Narasimhan et al., 2024)	hybrid	cascade deferral score	token / stage	handoff, stop	Mid

Table 3: Confidence-guided model-selection methods, organized by signal source, signal form, decision unit, functional role, and operational access.

Access: Pre = pre-call routing; Post = post-hoc deferral after a provisional answer; Mid = mid-generation hand-off; MS = multiple sampled outputs; Aux = auxiliary router, judge, or classifier; FT = extra post-training or confidence tuning. The table focuses on core routing and cascade methods. Adjacent orchestration methods such as MoA and token-stitching methods such as R-Stitch (Chen et al., 2025b) are discussed in the text but omitted here.

Many apparent disagreements between papers are better understood as consequences of this timing difference rather than as contradictions about which confidence signal is universally best.

The section also shows that confidence source becomes more heterogeneous once multiple models are involved. Some systems rely on self-signals from the cheaper model, including token uncertainty, learned confidence tokens, or semantic entropy. Others depend on external predictors, such as query-level performance models or optimization layers. A third group uses judge-style signals, including pairwise LLM judgments or multi-judge agreement, to estimate relative model quality. These sources are not interchangeable: self-signals are often cheaper at deployment once a candidate answer exists, whereas external and judge-based signals are more natural for pre-call routing or expert selection across larger model pools.

Finally, the strongest recent systems increasingly blur the boundary between routing and cascading. Unified-routing formulations and taxonomy-guided shortlist-then-escalate pipelines both suggest that practical systems rarely perform only one kind of confidence-guided decision. Instead, they layer ex-ante pruning, post-hoc checking, and selective escalation. This is precisely where the broader definition of confidence adopted in this survey is useful: the controlling signal may be token uncertainty, semantic entropy, judge agreement, predicted quality gap, or bootstrap uncertainty, but in each case it is being used to decide how the system allocates model capacity.

6 Confidence-Gated RAG Systems

RAG introduces confidence-guided control at several distinct stages of the same pipeline. In the notation of Section 2, the decision state now contains the query, any retrieved documents or snippets, partial generations, and, in some systems, verifier outputs or retrieval-quality estimates. The decision units therefore range from the query itself to documents, snippets, claims, and final answers. The corresponding actions are equally varied: decide whether retrieval is needed at all, choose a retrieval mode, filter or rerank context, verify groundedness after generation, or abstain when the resulting answer is not trustworthy enough.

What makes RAG distinctive is that confidence must be interpreted relative to *source*. A high-confidence answer may reflect strong parametric knowledge, strong support from retrieved evidence, or an unresolved conflict between the two. This is why the RAG literature repeatedly distinguishes internal knowledge from external support, and why diagnostic work such as Soudani et al. (2025) is so important: uncertainty methods that work in no-retrieval settings do not automatically transfer once retrieved context changes the model’s epistemic state. The section is organized accordingly, moving from retrieval gating to context selection, then to post-hoc groundedness assessment and abstention. Table 4 summarizes where confidence enters the RAG pipeline and what decision it governs at each stage.

6.1 When to Retrieve

The first RAG control question is whether external evidence is needed at all. Here the relevant confidence signal is usually a query-level or generation-level estimate of the marginal value of retrieval, not a final answer confidence in the ordinary sense. Some methods make this decision during generation. Jiang et al. (2023) trigger retrieval in FLARE when predicted upcoming content contains low-confidence tokens, using those predictions to form forward-looking retrieval queries. Su et al. (2024b) broaden this idea by modeling the model’s evolving information need rather than only reacting to a local token heuristic, and they explicitly improve both retrieval timing and retrieval query construction. In both cases, retrieval is activated when the ongoing generation suggests that parametric knowledge alone is no longer sufficient.

Other systems decide earlier, at the query level. Jeong et al. (2024) frame retrieval as strategy selection: the model should choose among no retrieval, single-step retrieval, and more elaborate iterative pipelines according to estimated question complexity. Wang et al. (2023b) use self-knowledge signals to decide whether a question lies within the model’s parametric competence, so retrieval is triggered only when the model judges that outside support is needed. Mallen et al. (2023) provide the broader motivation for this line of work by clarifying the trust boundary between parametric and non-parametric memory, even though their paper is more analytical than algorithmic.

Several recent methods make the control signal richer than a one-bit trigger. Asai et al. (2024) train SELF-RAG with reflection tokens that jointly govern whether to retrieve, whether retrieved passages are relevant and supportive, and whether a generated answer is useful. Yao et al. (2025) likewise use internal-state uncertainty as a multi-role signal that supports retrieval triggering, snippet reranking, and strategy choice. Li et al. (2025a) study temporal entropy patterns as another way to anticipate retrieval needs before errors fully materialize. Zubkova et al. (2025) extend adaptive gating to retrieval depth, using semantic uncertainty to choose not only whether to retrieve but also whether single-step or multi-step retrieval is warranted. Finally, Chen et al. (2025a) use agreement between two parametric answer paths as a verification signal: if the two paths converge, retrieval can be skipped; if they diverge, the system retrieves and then filters documents adaptively. Taken together, these papers show that “when to retrieve” is best understood as a decision about the expected value of non-parametric knowledge under the current state, not merely as thresholding a generic uncertainty score.

6.2 What Context to Keep

Once retrieval has been invoked, the next control problem is context allocation: which retrieved units are actually worth preserving for generation? This stage is different from retrieval gating because the relevant signal now concerns retrieval quality, passage utility, or snippet coherence rather than parametric self-knowledge. Yan et al. (2024) exemplify this shift. CRAG uses a retrieval evaluator to score the quality of the retrieved set and then branches to different corrective actions: keep and refine the context when retrieval is good, fall back to web search when it is poor, or mix strategies when confidence is ambiguous. The signal therefore controls *how* to repair retrieval, not only whether retrieval should have happened.

Other papers make passage utility itself the scored object. Wang et al. (2025b) define Document Information Gain as the change in generation confidence caused by a document and use that signal for reranking and filtering. This is one of the clearest document-level confidence-utilization ideas in the RAG literature because the score is explicitly answer-conditioned rather than based only on semantic similarity. Wang et al. (2023c) similarly treat context filtering as a first-class learned decision point before generation. Isoda (2025) push filtering to sentence level by using elicited self-knowledge as a signal for which retrieved content should be retained, while Zhu et al. (2024) integrate document selection into decoding itself through sparse context selection. In a broader adaptive-RAG view, both SELF-RAG and SEAKR also belong here because their reflection or uncertainty signals are reused after retrieval to critique and prioritize external evidence rather than merely to trigger retrieval.

Long-context retrieval introduces another unit of control: the span or chunk. Li et al. (2024e) show that chunk boundaries can destabilize retrieval and therefore model span-level uncertainty directly to improve similarity estimation and robustness under chunking noise. This is an important reminder that RAG confidence is not

Method	Source	Signal	Unit	Role	Access
When to retrieve					
FLARE (Jiang et al., 2023)	self	low-confidence future tokens	sentence / token	trigger, regenerate	Mid
DRAGIN (Su et al., 2024b)	self	information-need uncertainty	context / step	trigger, query	Mid
Adaptive-RAG (Jeong et al., 2024)	ext	predicted question complexity	query	route strategy	Pre, FT
SKR (Wang et al., 2023b)	self	self-knowledge elicitation	query	retrieve-or-skip	Pre
SELF-RAG (Asai et al., 2024)	self	reflection tokens	query / passage / answer	trigger, critique	Pre, Ctx, FT
SEAKR (Yao et al., 2025)	mech	self-aware internal uncertainty	query / snippet / strategy	trigger, rerank, route	Pre, Ctx, WB
SUGAR (Zubkova et al., 2025)	self	semantic uncertainty	query	trigger, choose depth	Pre, MS
PAIRS (Chen et al., 2025a)	self	parametric-path convergence	query / document	trigger, filter	Pre, Ctx
What Context to Keep					
CRAG (Yan et al., 2024)	ext	retrieval-quality score	retrieval set	correct, fallback	Ctx, Aux
FILCO (Wang et al., 2023c)	ext	context usefulness score	passage / sentence	filter	Ctx, FT
InfoGain-RAG (Wang et al., 2025b)	self	document information gain	document	rerank, filter	Ctx
SKILL-RAG (Isoda, 2025)	self	self-knowledge sentence score	sentence	filter	Ctx, FT
Sparse-RAG (Zhu et al., 2024)	self	sparse document-selection score	document	select, sparsify	Ctx, Mid
UncertaintyRAG (Li et al., 2024e)	self	span uncertainty / SNR	span / chunk	score, retrieve	Ctx, FT
Groundedness and Hallucination Detection					
ReDeEP (Sun et al., 2024)	mech	ECS + PKS	answer / mechanism	detect, mitigate	Post, WB
HALT-RAG (Goswami & Kurra, 2025)	ext	calibrated NLI ensemble	claim / answer	detect, abstain	Post, Aux
FRANQ (Fadeeva et al., 2025)	ext	faithfulness-conditioned factuality UQ	claim	detect	Post, Aux
Abstention and Conformal Guarantees					
TRAQ (Li et al., 2024d)	hybrid	conformal passage + answer scores	passage set / answer set	set-predict	Ctx, Post, CP
ConFLARE (Rouzrokh et al., 2024)	ext	conformal similarity threshold	chunk / retrieval set	calibrate retrieval	Ctx, CP
Principled Ctx Eng (Chakraborty et al., 2025)	ext	conformal snippet relevance	snippet	filter	Ctx, CP
Conformal-RAG (Feng et al., 2025)	hybrid	conditional conformal factuality	sub-claim	filter	Post, CP
Divide-Then-Align (Sun et al., 2025)	hybrid	knowledge-boundary quadrant	query	abstain, align	Post, FT

Table 4: Confidence-guided RAG methods, organized by signal source, signal form, decision unit, functional role, and operational access.

Access: Pre = before retrieval; Mid = during generation or active retrieval; Ctx = retrieved-context scoring or filtering stage; Post = after answer generation; MS = multiple samples; WB = white-box activations or logits; Aux = auxiliary evaluator, verifier, or classifier; FT = extra post-training or learned controller; CP = conformal calibration set / formal coverage guarantee. Background, dataset, and diagnostic papers such as Mallen et al. (2023); Niu et al. (2024); Lymperopoulos & Sarathy (2025); Soudani et al. (2025); Tao et al. (2024) are discussed in the text but omitted here.

only about the final answer. It can also attach to passages, snippets, or spans whose estimated usefulness or stability determines how much external context the generator actually sees.

6.3 Groundedness and Hallucination Detection

Even with retrieved evidence, RAG systems can still hallucinate by ignoring context, overtrusting parametric knowledge, or mixing the two sources improperly. At this stage the central question is no longer whether to retrieve or which evidence to keep, but whether the generated content is actually grounded in the available evidence. The field has benefited from stronger evaluation resources here. Niu et al. (2024) provide a fine-grained hallucination corpus for RAG, but the paper should be understood as dataset infrastructure rather than as a direct confidence-utilization method. Its main value in this survey is that it enabled later groundedness detectors to be studied on realistic RAG errors.

Among those detectors, Sun et al. (2024) are especially important because they make the source conflict explicit. ReDeEP uses mechanistic signals to separate parametric knowledge use from external-evidence use, showing that hallucination often arises when internal parametric mechanisms dominate despite available context. Goswami & Kurra (2025) take a more black-box route and use calibrated NLI-ensemble features for post-hoc hallucination detection with abstention. Fadeeva et al. (2025) sharpen the target further by distinguishing factuality from faithfulness: a statement can be unsupported by the retrieved context without being false in the world, and a good uncertainty estimate should not collapse those cases together. This distinction is one of the most important conceptual corrections in the RAG literature.

Several adjacent papers broaden the notion of source-aware confidence. Chen et al. (2025a) compare parametric and retrieval-informed answer paths, using their convergence or divergence as a signal for whether external evidence has changed the system’s belief meaningfully. Lymperopoulos & Sarathy (2025) extend uncertainty estimation to tool-augmented systems more generally, including RAG as one application, by jointly modeling LLM and tool uncertainty. Soudani et al. (2025) then provide the key diagnostic result for the whole section: uncertainty estimators that look reasonable in plain LLM settings often violate basic desiderata once retrieved

context is introduced. Their contribution is not a new adaptive RAG pipeline, but a principled warning that RAG-specific uncertainty should be evaluated differently from no-retrieval uncertainty.

6.4 Abstention and Conformal Guarantees

The final RAG control problem is what to do when confidence remains insufficient even after retrieval and post-hoc checking. One answer is abstention; another is to return calibrated sets or filtered claims with formal guarantees. The conformal line of work is especially important here because different papers guarantee different objects. Li et al. (2024d) apply conformal prediction to both retrieval and generation, producing passage sets and answer sets with end-to-end correctness guarantees at the set level. Rouzrokh et al. (2024) focus more narrowly on retrieval coverage by calibrating similarity thresholds so that answer-containing chunks are included with target confidence. Chakraborty et al. (2025) move the guarantee to snippet filtering before generation, while Feng et al. (2025) apply conditional conformal prediction at the sub-claim level to filter generated claims according to factuality guarantees. These papers belong together, but they should not be treated as interchangeable because they calibrate different units and guarantee different notions of success.

Abstention can also be learned rather than fully conformalized. Sun et al. (2025) model the joint knowledge boundary of parametric and retrieved knowledge and train the system to refuse when neither source is adequate. This is a confidence-utilization method in a broad sense, but its signal is knowledge-boundary membership rather than a post-hoc scalar threshold. HALT-RAG, discussed above, sits nearby as a practical precision-constrained abstention system built on calibrated verification scores. These methods show that trustworthy RAG often depends as much on deciding *not* to answer as on retrieving the right evidence.

More broadly, confidence-alignment work such as Tao et al. (2024) is relevant to RAG because it prepares verbalized confidence for downstream trust and retrieval decisions. However, it is best read as a bridge from confidence elicitation to RAG control rather than as a core RAG pipeline paper. That distinction matters for the survey: some methods are native RAG mechanisms, while others supply better confidence signals that RAG systems can later consume.

Discussion. Across these subsections, RAG makes confidence more source-sensitive than any earlier stage of the lifecycle. Before retrieval, the signal estimates whether external evidence is worth paying for. After retrieval, the signal estimates which evidence is useful enough to retain. After generation, the signal estimates whether the resulting content is grounded, faithful, or safe enough to return. The same general pattern from Section 2 still holds, but the unit and the meaning of the score change sharply from one stage to the next.

This is why source-aware distinctions are essential in RAG. Parametric self-knowledge, retrieval-quality estimates, document utility scores, mechanistic signals, NLI-based support scores, and conformal nonconformity scores are all confidence-like control signals, but they answer different questions. In particular, the literature now makes clear that faithfulness and factuality must be separated, and that confidence in internal knowledge must be distinguished from confidence grounded in retrieved context. Papers such as FRANQ, ReDeEP, PAIRS, and the axiomatic analysis of Soudani et al. (2025) are especially valuable because they expose this distinction directly rather than assuming a single scalar uncertainty can cover all cases.

The open problem is therefore not merely better calibration of one score. It is how to represent and use confidence in a way that tracks *where* the model’s belief comes from. RAG has already established that confidence can gate retrieval, filter context, verify groundedness, and support abstention or formal guarantees. The next frontier is source-aware confidence that explicitly models the interaction between parametric memory and contextual evidence instead of treating them as interchangeable contributors to one undifferentiated belief state.

7 Confidence-Based Risk Management

Risk management is the point where confidence becomes an explicit decision variable. Given a candidate answer, a set of atomic claims, or a set-valued prediction, the system may abstain, trigger verification, flag hallucination risk, or return a calibrated set rather than a single output. In the notation of Section 2, the decision units in this section are primarily answers, claims, and prediction sets, and the downstream actions

are DETECT, ABSTAIN, COVER, and CALIBRATE. The emphasis is therefore different from earlier sections: the goal is not merely to estimate confidence, but to use it to reduce failure in settings where unsupported answers are costly.

Adjacent surveys cover parts of this landscape from different angles. Rawte et al. (2023) review hallucination across foundation models, while Wen et al. (2025) survey abstention in large language models. Our focus here is narrower and more action-oriented: how confidence-like signals are obtained, and then converted into concrete risk-control decisions. To keep that hierarchy clear, we treat confidence elicitation as an enabling layer, and then examine three main downstream families: hallucination and failure detection, conformal coverage, and abstention-oriented reliability alignment. Table 5 groups the risk-management literature by signal form, protected unit, and operational guarantee.

7.1 Obtaining Actionable Confidence Signals

Before confidence can gate risk, systems need access to some actionable signal about answer reliability. Earlier sections consume confidence primarily for efficiency or accuracy, but risk management places stronger demands on signal quality, such as capturing directional overconfidence, remaining meaningful under input ambiguity, and generalizing across evaluation contexts. Foundational work introduced self-knowledge formulations such as $P(\text{True})$ and $P(\text{IK})$, separating confidence in a proposed answer from confidence that the model knows the answer at all (Kadavath et al., 2022). Closely related work evaluates self-knowledge more directly through the ability to recognize unanswerable or unknowable questions, framing uncertainty as awareness of the model’s own knowledge boundary rather than only confidence in a proposed answer (Yin et al., 2023). Later work showed that, especially for RLHF models, prompted verbal or numerical confidence can be better calibrated than raw conditional probabilities, particularly when the model is asked to consider alternative answers before committing to a score (Tian et al., 2023). These signals matter in the present section not because elicitation is the end goal, but because abstention, selective generation, and post-hoc verification all depend on having some usable estimate in the first place.

However, the usefulness of an elicited signal depends on what kind of uncertainty is present. Cole et al. (2023) show that abstention is not only about epistemic ignorance; denotational ambiguity creates a separate failure mode in which the model answers confidently even though the question itself is underspecified. This is one reason why scalar calibration metrics alone are incomplete. Subsequent work sharpens this diagnosis from several directions: Net Calibration Error makes directional overconfidence explicit (Groot & Valdenegro Toro, 2024); verbal uncertainty can be traced to a representation-space feature and adjusted through mechanistic intervention (Ji et al., 2025); and consistency under uncertain-expression prompting provides a cheap black-box factuality signal without logit access (Joo et al., 2025). Even so, elicited confidence remains fragile. Verbal confidence can help in some answering settings (Tian et al., 2023), yet it remains overconfident in evaluator pipelines such as LLM-as-a-Judge (Tian et al., 2025). For that reason, the remainder of this section treats elicitation as a prerequisite for risk control rather than as an end in itself.

7.2 Hallucination and Failure Detection

One major use of confidence in risk management is to identify outputs whose factuality is doubtful before they are acted upon downstream. Black-box methods exploit behavioral instability. Manakul et al. (2023) use divergence across sampled responses as a zero-resource hallucination signal, showing that self-consistency alone can separate factual from non-factual statements. Farquhar et al. (2024) replace raw lexical disagreement with semantic entropy, clustering sampled answers by meaning and measuring uncertainty over semantic clusters rather than surface forms. This shift is important because many incorrect outputs differ only lexically, whereas semantic entropy targets confabulations at the level of meaning.

Later work compresses or redirects these signals for deployment. Han et al. (2024) approximate semantic entropy with cheap probes over hidden states from a single generation, preserving much of the original detector’s strength while removing repeated sampling at test time. Chen et al. (2024a) and Azaria & Mitchell (2023) instead access internal representations directly, using hidden-state classifiers or semantic-space consistency scores to detect false or overconfident generations even when the decoded text looks fluent. At the black-box end, Joo et al. (2025) use consistency under uncertain-expression prompting as a lower-

Method	Source	Signal	Unit	Role	Access
Actionable Signals					
P(True) / P(IK) (Kadavath et al., 2022)	self	self-knowledge score	answer / query	elicit	Pr
Just Ask (Tian et al., 2023)	self	verbal / numerical confidence	answer	elicit, calibrate	Pr
VUF / MUC (Ji et al., 2025)	mech	verbal-semantic mismatch	answer / hidden state	detect, steer	WB
UExpr Consistency (Joo et al., 2025)	self	uncertain-expression consistency	answer	detect	BB
Hallucination / Failure Detection					
SelfCheckGPT (Manakul et al., 2023)	self	inter-sample consistency	sentence / answer	detect	BB, MS
Semantic Entropy (Farquhar et al., 2024)	self	entropy over meaning clusters	answer set	detect	MS
SEPs (Han et al., 2024)	mech	probed semantic entropy	answer / hidden state	detect	WB
INSIDE (Chen et al., 2024a)	mech	EigenScore + clipping	answer / hidden state	detect	WB
Internal State (Azaria & Mitchell, 2023)	mech	hidden-state truth probe	statement / answer	detect	WB
Conformal Guarantees					
MCQA CP (Kumar et al., 2023)	self	option nonconformity	answer set	set-predict	CP
Conformal LM (Quach et al., 2024)	self	sampling + rejection score	output set	set-predict, filter	MS, CP
Conformal Factuality (Mohri & Hashimoto, 2024)	hybrid	entailment-based nonconformity	claim	filter, back off	CP
Enhanced CP (Cherian et al., 2024)	hybrid	conditional factuality score	claim	filter	CP
Coherent Factuality (Rubin-Toles et al., 2025)	hybrid	graph-structured conformal score	claim graph	filter, preserve coherence	CP
API-Only CP (Su et al., 2024a)	hybrid	sample frequency + semantic similarity	answer set	set-predict	API, MS, CP
ConfTS (Xi et al., 2025)	self	conformal temperature scaling	prediction set	calibrate	CP
Abstention / Reliability Alignment					
Self-Evaluation (Ren et al., 2023)	self	token-level self-eval	answer	abstain, select	Pr
Adapt w/ Self-Eval (Chen et al., 2023a)	self	adapted self-eval score	answer	abstain	PE
Selective Ambig. QA (Cole et al., 2023)	self	repetition / agreement	query / answer set	abstain	MS
R-Tuning (Zhang et al., 2024b)	self	parametric knowledge boundary	query / answer	abstain, align	FT
SemEnt FT (Tjandra et al., 2024)	self	semantic entropy	answer	abstain, align	FT, MS
ConfTuner (Li et al., 2025e)	self	tokenized Brier verbal confidence	answer	calibrate	FT
FISCORE (An & Xu, 2025)	self	semantic cluster consensus	answer sample	abstain, align	RL, MS
Act.-Based Abst. (Huang et al., 2025d)	mech	FFN activation confidence	answer	abstain	WB

Table 5: Confidence-based risk-management methods, organized by signal source, signal form, decision unit, functional role, and operational access.

Access: Pr = prompted elicitation or self-evaluation; BB = black-box prompting only; WB = white-box hidden states or logits; MS = multiple samples; API = API-only deployment; PE = parameter-efficient adaptation; FT = extra fine-tuning; RL = reinforcement learning; CP = conformal guarantee. Diagnostic metrics, surveys, benchmarks, and adjacent papers such as NCE, LLM-as-a-Judge overconfidence, FACTSCORE, SafeBehavior (Zhao et al., 2025), Li et al. (2025b), Tripathi et al. (2025), and Wen et al. (2025) are discussed in the text but omitted here.

cost factuality cue, while Muhammed et al. (2025) show that many of these detectors transfer unevenly across domains, especially from biography-style factuality to mathematical reasoning. In long-form settings, frameworks such as FACTSCORE provide the atomic-claim decomposition that many later detection and conformal methods rely on, even though they are primarily evaluation infrastructure rather than detectors themselves (Min et al., 2023).

7.3 Conformal Prediction and Coverage Guarantees

Conformal methods use confidence or nonconformity scores in a different way. Rather than ranking answers by trust alone, they convert uncertainty into set-valued outputs or filtered responses with explicit coverage guarantees. Early LLM work applies split conformal prediction to multiple-choice QA, producing answer sets whose size reflects uncertainty and whose guarantees depend on the usual exchangeability assumptions (Kumar et al., 2023). For open-ended generation, conformal language modeling calibrates sampling and rejection rules over candidate outputs (Quach et al., 2024), while API-only methods replace logits with sampling frequency and semantic similarity so that conformal prediction remains possible in black-box deployments (Su et al., 2024a).

Another line targets factuality at the claim level. Mohri & Hashimoto (2024) introduce conformal factuality, progressively backing off from a response by removing the least certain claims until the remaining output satisfies a target correctness level. Subsequent work strengthens this family in two directions: Cherian et al. (2024) make the guarantees more adaptive and topic-sensitive, and Rubin-Toles et al. (2025) extend conformal factuality to reasoning tasks where claims cannot be filtered independently but must remain coherent as a graph. A key lesson from this literature is that conventional calibration and conformal efficiency are not the same objective. Standard calibration procedures can enlarge conformal sets, which is why Xi et al. (2025) optimize directly for conformal efficiency rather than ordinary ECE. General conformal risk control provides the theoretical backdrop, but the practical design choice in LLMs is the unit to which the guarantee is attached: answer options, sampled outputs, or atomic claims (Angelopoulos et al., 2024). Relatedly, Gui

et al. (2024) treat trustworthiness itself as the conformal target, using a calibrated threshold to select outputs whose predicted alignment scores imply a user-specified reliability level. A complementary recent direction, Conformal Linguistic Calibration, reinterprets linguistically hedged answers as answer-set prediction and uses conformal guarantees to trade off factuality against specificity, thereby linking coverage control to verbal uncertainty expression more directly (Jiang et al., 2025).

7.4 Abstention and Reliability Alignment

Abstention methods use confidence more directly: the model answers when the signal suggests adequate support and refuses, defers, or hedges when it does not. In selective generation, token-level self-evaluation often outperforms raw sequence likelihood for deciding when to abstain on free-form outputs (Ren et al., 2023), and parameter-efficient adaptation can improve the quality of those self-evaluation signals for task-specific selective prediction (Chen et al., 2023a). R-Tuning turns this into a training objective by constructing refusal-aware supervision around the model’s parametric knowledge boundary (Zhang et al., 2024b), while semantic-entropy fine-tuning provides a label-free alternative that extends more naturally to long-form generation (Tjandra et al., 2024).

More recent work refines the abstention signal itself. Huang et al. (2025d) use activation-based uncertainty estimation to support low-latency response abstinence in high-stakes RAG deployments, Li et al. (2025e) train models to verbalize calibrated confidence through a tokenized Brier objective, and An & Xu (2025) use fine-grained semantic consensus as a reward for abstention-oriented reinforcement learning. These methods are best understood as reliability alignment rather than pure calibration: they attempt to move the refusal boundary itself, not merely to post-process a fixed score. That distinction is central in the abstention literature surveyed by Wen et al. (2025), which emphasizes that query answerability, model confidence, and value alignment can all justify refusal. It is also why robustness matters. Verbal confidence may be useful under nominal prompting, yet it remains vulnerable to adversarial manipulation (Obadinma & Zhu, 2025), and broader work on calibration attacks suggests that confidence robustness is itself a separate design objective rather than a by-product of nominal calibration (Obadinma et al., 2024). We return to this issue in Section 9 (Challenge 5), where we discuss robustness and portability as a cross-cutting open problem.

Discussion. The preceding subsections suggest that confidence-based risk management serves at least three distinct objectives. Calibration-oriented work asks whether a signal tracks empirical correctness or uncertainty. Selective-reliability work asks whether the system abstains, filters, or escalates at the right boundary. Conformal work asks whether set-valued or filtered outputs achieve a target error rate while remaining useful. These objectives are related, but they are not interchangeable, and the literature becomes much easier to read once that distinction is made explicit.

This perspective also explains several apparent tensions in the literature. Verbalized confidence can outperform raw probabilities in some RLHF answering settings (Tian et al., 2023) yet remain overconfident in evaluator pipelines or under adversarial prompting (Tian et al., 2025; Obadinma & Zhu, 2025). Calibration-focused fine-tuning can improve verbal honesty or ECE (Li et al., 2025e) without improving conformal efficiency (Xi et al., 2025). Likewise, methods that reshape the refusal boundary through tuning or reinforcement learning (Zhang et al., 2024b; Tjandra et al., 2024; An & Xu, 2025) solve a different problem from post-hoc detectors such as semantic entropy or SelfCheckGPT (Farquhar et al., 2024; Manakul et al., 2023). The practical question is therefore not which method is “most calibrated” in the abstract, but which guarantee the deployment setting actually requires.

In practice, disagreement- and representation-based detectors are appropriate when the goal is to warn on likely hallucinations; abstention-alignment methods are more suitable when the system must refuse unsupported answers; and conformal methods are the right tool when auditable coverage guarantees are required, even at the cost of larger sets or less specific outputs. Across all three settings, the open problem is joint design: obtaining confidence signals that remain informative under ambiguity, robust under attack, and useful for downstream control rather than only for post-hoc reporting.

8 Confidence in Agentic Systems

Confidence-guided agentic control extends the survey from single decisions to composed action loops. In agent settings, the system must decide not only which answer to trust, but whether to call a tool, allocate more compute, revise a partial plan, backtrack from an error state, or aggregate multiple agents into a final action. Under the broad definition used throughout this survey, the relevant signals therefore include self-confidence, verifier uncertainty, process-reward estimates, hesitation features, and communicated peer confidence. The common pattern from Section 2 still applies: a reliability-relevant score is attached to a decision unit and then converted into an action. What changes is that the unit itself can now be a tool call, a reasoning step, a search branch, a full trajectory, or a team-level vote.

Extensions to the core notation. Agentic control benefits from two additional descriptors. The first is *temporal horizon*: confidence may attach to a local step, an intermediate branch, a full trajectory, or an end-to-end episode. The second is *actor scope*: the relevant signal may come from the acting agent itself, from a verifier or reward model, from peer agents in a discussion, or from an external orchestrator. In practice, the state ξ_t now contains not only the prompt and partial output, but also environment observations, memory, tool outputs, and peer messages. This section uses those two descriptors to organize the literature into four layers of control: selective escalation, self-correction, verifier-guided search, and multi-agent aggregation. Table 6 collects the core agentic methods and highlights which signals drive escalation, search, or aggregation.

8.1 Selective Escalation, Tool Use, and Budget Allocation

The first agentic control problem is whether additional action is warranted at all. Rather than always debating, always searching, or always invoking more compute, several recent systems use confidence-like signals to decide when escalation is likely to pay off. Fan et al. (2025) study this question directly for multi-agent debate. Instead of triggering debate through a raw confidence threshold, they extract structured hesitation features from a self-critique and train a lightweight classifier to decide whether debate is likely to improve the answer. The signal is therefore query-level and policy-facing: invoke a more expensive collaborative procedure only when the expected gain justifies the cost.

Adaptive compute allocation makes the same decision at a different granularity. Park et al. (2025) calibrate process reward model (PRM) success estimates so that they can be used as decision-grade confidence signals for allocating additional search or best-of- N budget. Snell et al. (2025) frame the broader systems problem: inference-time compute should be allocated unevenly across prompts, since easy queries benefit less from extra search than medium-difficulty ones. In both cases, the decision is not which answer is currently best, but whether a partial trajectory or prompt deserves more budget. Verification-aware planning systems such as Xu et al. (2026) push this logic into coordination itself, using planner-defined verification functions to decide whether a subtask should proceed, retry, or trigger replanning.

This selective-escalation view also clarifies what does *not* belong centrally in this section. Confidence-gated edge-cloud routing, dynamic retrieval, and answer rejection are closely related ideas, but papers such as Zhang et al. (2025c); Chuang et al. (2025b); Jin et al. (2025); Chuang et al. (2025a); Fu et al. (2025) are more naturally treated in Sections 5, 6, and 4, where the controlled units are models, retrieved context, or candidate answers rather than native agent actions.

8.2 Self-Correction, Revision, and Backtracking

The next control layer operates inside an ongoing trajectory. Foundational baselines such as Madaan et al. (2023) and Shinn et al. (2023) show that language feedback can improve future behavior, either within a single critique-and-revise loop or across repeated trials with memory. These papers are essential context, but their signals are mostly holistic and verbal: the model critiques an output or stores a lesson, rather than explicitly deciding which local part of a trajectory is unreliable.

Several later papers make that decision much more explicit. Lee et al. (2025) learn intrinsic self-verification and an internal stop-versus-revise decision, then use the resulting verification confidence both to determine whether refinement should continue and to bias inference-time decoding. Shi et al. (2025) move from whole-answer critique to process-level repair by decomposing reasoning into Socratic steps, estimating confidence

Method	Source	Signal	Unit	Role	Access
Selective Escalation					
iMAD (Fan et al., 2025)	self	hesitation-feature trigger	query	trigger debate	Pr, FT, MA
PRM Calibration (Park et al., 2025)	auxiliary	calibrated PRM success	prefix / traj	allocate budget	Aux, MS, FT
Scaling TTC (Snell et al., 2025)	hybrid	difficulty / verifier value	query / traj	allocate compute	Aux, MS
VeriMAP (Xu et al., 2026)	auxiliary	planner verification funcs	subtask / plan node	retry, replan	Env, Aux
Self-Correction / Backtracking					
ReVISE (Lee et al., 2025)	self	intrinsic self-verification	step / traj	revise, stop	FT
SSR (Shi et al., 2025)	self	step self-consistency	Socratic step	locate, refine	MS, Pr
BacktrackAgent (Wu et al., 2025c)	auxiliary	verifier + judge score	page / action state	detect, backtrack	Env, Aux
Verifier- and Reward-Guided Search					
LATS (Zhou et al., 2024)	hybrid	value + reflection signal	node / traj	expand, select	Env, MS
ReST-MCTS (Zhang et al., 2024a)	auxiliary	inferred process reward	step / traj	expand, filter	MS, FT
UATS (Song et al., 2026)	auxiliary	PRM uncertainty (MC Dropout)	step / node	guide search, allocate budget	Aux, MS, FT
AgentRM (Xia et al., 2025)	auxiliary	trajectory reward score	step / traj	guide search	Aux, FT
RewardAgent (Peng et al., 2025)	hybrid	RM + verifier reward	response / traj	select, supervise	Aux, FT
Multi-Agent Deliberation					
ReConcile (Chen et al., 2024b)	peer	agent confidence + explanations	answer / round	aggregate, update	MA
ConfMAD (Lin & Hooi, 2025)	peer	LN / verbal confidence	response / turn	aggregate, revise	MA

Table 6: Confidence-guided agentic control methods, organized by signal source, signal form, decision unit, functional role, and operational access.

Access: Pr = prompted self-critique or self-evaluation; Env = interactive environment or planner-mediated execution; Aux = external verifier, reward model, or planner module; MS = multiple samples / rollouts; FT = extra post-training or learned controller; MA = multi-agent discussion. Baseline or diagnostic papers such as Madaan et al. (2023); Shinn et al. (2023); Kim et al. (2025); Yang et al. (2025); Cui et al. (2025); Kaesberg et al. (2025); Wu et al. (2025a) and off-section routing / RAG papers such as Zhang et al. (2025c); Chuang et al. (2025b); Jin et al. (2025); Chuang et al. (2025a); Fu et al. (2025) are discussed in the text but omitted here.

for each step through repeated sub-question solving, and revising from the weakest point. In interactive environments, Wu et al. (2025c) use verifier and judge signals to decide whether the current GUI state indicates an error that warrants backtracking. What these papers share is not merely self-improvement, but explicit local control: confidence or verification is attached to a step or state, and that signal determines whether the agent should continue, revise, or roll back.

Other agentic papers are better understood as *behavioral* rather than *calibration-native* control. Kim et al. (2025) teach reflection and backtracking from search-derived traces, while Yang et al. (2025) explicitly train models to emit backtracking actions during reasoning. These methods clearly govern downstream actions, but the control signal is embedded in learned search behavior rather than surfaced as a calibrated scalar. Distinguishing these cases matters for the survey: they are relevant agentic control papers, but they should not be presented as if they solve the same problem as intrinsic self-verification methods such as ReVISE or step-confidence methods such as SSR.

8.3 Verifier- and Reward-Guided Search

Search-based agent systems attach control signals to prefixes, nodes, or trajectories rather than final answers. Zhou et al. (2024) combine MCTS with reflections and environment feedback so that expansion decisions depend on value-like estimates over future trajectories. Zhang et al. (2024a) similarly use inferred process rewards to guide tree search and to filter self-training traces, treating partial reasoning quality as a signal for which branches merit further exploration. These are broad control frameworks rather than narrow confidence-estimation papers, but under the survey’s operational definition they belong here because the reward or value signal directly governs search actions.

The more confidence-native part of this literature asks whether the controller itself can be trusted. Song et al. (2026) quantify epistemic uncertainty in process reward models via Monte Carlo Dropout, showing that PRMs can be overconfident on out-of-distribution reasoning paths, and propose an RL-based controller that dynamically allocates search budget based on that uncertainty. Park et al. (2025) make the same issue operational by calibrating PRM success estimates before using them for adaptive inference-time scaling. Xia et al. (2025) and Peng et al. (2025) extend this logic to learned reward systems, where trajectory-level or response-level reward signals are used for best-of- N selection, beam search, or preference construction.

Across these papers, the recurring lesson is that once an agent’s search policy depends on a verifier or reward estimate, uncertainty in the verifier becomes part of the control problem itself.

8.4 Multi-Agent Deliberation and Aggregation

In multi-agent systems, confidence becomes a communicated social signal rather than a purely internal estimate. Chen et al. (2024b) combine diverse models, per-agent confidence, and calibrated confidence-weighted voting, arguing that debate works best when confidence is paired with model diversity and convincing corrective explanations. Lin & Hooi (2025) make this dependence even more explicit by comparing token-probability-derived and self-verbalized confidence, then studying how calibration changes debate quality. Fan et al. (2025) move one step earlier and ask whether debate should happen at all, using self-critique-derived hesitation features to trigger multi-agent deliberation only when it is expected to help.

At the same time, the strongest diagnostic papers caution against simplistic narratives. Wu et al. (2025a) show in a controlled setting that diversity and intrinsic reasoning strength matter more than visible confidence alone, and that debate can fail through majority pressure even when confidence is exposed. Kaesberg et al. (2025) and Cui et al. (2025) further show that aggregation protocol is not an implementation detail: task type, conformity pressure, and trajectory-level scoring all change whether collaboration helps or harms. The practical lesson is that confidence in debate is only useful when the surrounding protocol knows how to interpret it. Calibrated local confidence does not automatically imply calibrated collective behavior.

Discussion. Agentic control extends confidence utilization from one-step decisions to layered control policies over time. The key distinctions are temporal horizon, actor scope, and signal type: some methods rely on self-verification, some on verifier uncertainty or reward models, and some on communicated peer confidence. These distinctions clarify why papers such as ReVISE, SSR, UATS, PRM Calibration, ReConcile, and ConfMAD are core examples of confidence-guided control, whereas papers such as Self-Refine, Reflexion, LATs, or Free-MAD are better understood as baselines or adjacent control frameworks with weaker or more implicit confidence semantics.

The main open problem is composition. Current work typically calibrates one layer at a time: a debate trigger, a step verifier, a PRM, or a confidence-weighted vote. Far less is known about how these signals should interact when an agent first decides whether to deliberate, then searches with a verifier, and finally aggregates multiple proposals. Reliable agentic systems will require confidence and verification signals that remain meaningful as they propagate across tools, steps, trajectories, and interacting agents.

9 Open Challenges

Section 2 framed confidence utilization as a decision process in which a reliability-relevant signal $\kappa_t(u; \xi_t)$ is attached to a unit u under state ξ_t , optionally transformed, and then consumed by a policy that changes the system’s behavior. Across training, inference, routing, retrieval, risk control, and agentic systems, this abstraction proved broad enough to cover log-probability-derived scores, uncertainty estimates, agreement statistics, verbalized confidence, verifier outputs, reward-model scores, and peer-reported confidence. At the same time, the survey makes clear that the field still lacks a mature theory of how such signals should be interpreted, compared, and deployed. As shown in Fig. 4, we highlight five open challenges that recur across the full lifecycle.

Challenge 1: Heterogeneous confidence semantics. The survey deliberately uses “confidence” in a broad operational sense, but that breadth exposes a foundational problem: many useful control signals do not estimate the same underlying quantity. Verbalized confidence can outperform raw conditional probabilities for RLHF-tuned models (Tian et al., 2023); agreement across sampled reasoning paths can act as a practical reliability proxy (Wang et al., 2023a); semantic entropy measures uncertainty over meanings rather than strings (Farquhar et al., 2024); calibrated process reward models estimate the future success probability of a partial trajectory (Park et al., 2025); and debate systems surface peer confidence as a communication signal rather than a private scalar (Chen et al., 2024b; Lin & Hooi, 2025). All of these signals are operationally useful, yet they are not obviously commensurate. A central open problem is therefore semantic: when two

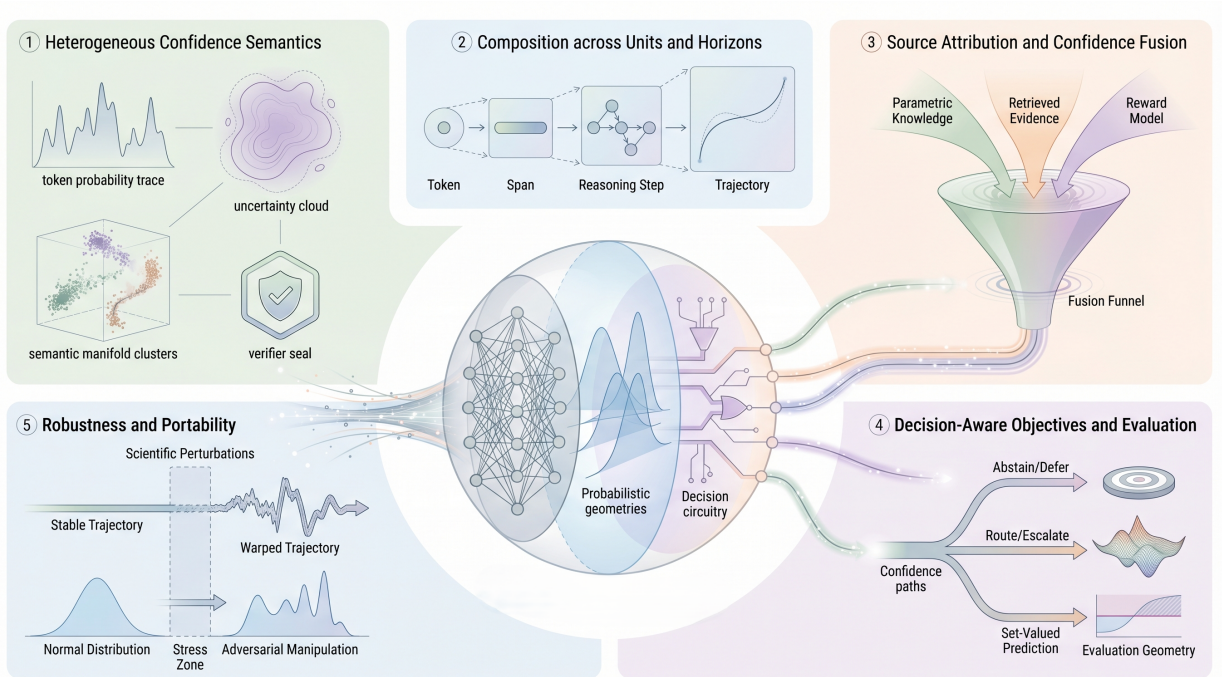


Figure 4: Conceptual illustration of the open challenges in confidence-guided control for large language models. The figure summarizes five recurring challenge areas identified in this survey: heterogeneous confidence semantics, composition across units and horizons, source attribution and confidence fusion, decision-aware objectives and evaluation, and robustness and portability.

methods expose different confidence signals, when should those signals be interpreted as competing estimates of the same property, and when do they represent fundamentally different objects?

Challenge 2: Composition across units and horizons. Confidence utilization rarely ends at the level where the signal is first computed. Local signals must usually be lifted into broader decisions: low-confidence token groups can trigger regeneration or early stopping (Fu et al., 2025), weak reasoning steps can trigger local repair (Shi et al., 2025), calibrated prefix-success estimates can decide how much additional search a trajectory deserves (Park et al., 2025), and confidence-weighted votes can decide the outcome of multi-agent discussion (Chen et al., 2024b). Yet there is still no general principle for propagating reliability from tokens to answers, from steps to trajectories, or from individual agents to collective action. Existing systems typically rely on heuristic aggregation, repeated sampling, or pipeline-specific controllers. A major open direction is to develop principled composition rules for turning local confidence into global control policies without losing the information that made the local signal useful in the first place.

Challenge 3: Source attribution and confidence fusion. The strongest systems increasingly combine several confidence sources at once, but they rarely preserve the provenance of those signals. This is especially visible in retrieval-augmented generation, where ReDeEP shows that parametric knowledge and retrieved evidence can disagree internally, and that accurate diagnosis may require disentangling the two sources mechanistically (Sun et al., 2024). Similar source-allocation problems appear elsewhere: routing systems may combine self-reported confidence with external routers (Chuang et al., 2025a); adaptive inference may rely on verifier or reward-model scores whose own uncertainty must be modeled (Park et al., 2025); and debate systems expose peer confidence that can help or harm depending on how the protocol interprets it (Lin & Hooi, 2025). The open challenge is not just to fuse more signals, but to build provenance-aware confidence representations that keep track of where a belief comes from, detect when sources conflict, and determine which source should dominate which downstream decision.

Challenge 4: Decision-aware objectives and evaluation. One of the clearest lessons of this survey is that confidence quality cannot be summarized by a single metric. A signal that is useful for abstention is not necessarily optimal for conformal prediction; a signal that is well calibrated in isolation may still be poor for routing or adaptive compute allocation. Cost-aware routing systems optimize the quality-cost frontier rather than classical calibration (Chen et al., 2023b); conformal methods optimize coverage and set efficiency rather than ordinary pointwise confidence (Quach et al., 2024; Xi et al., 2025); abstention-oriented training shifts the refusal boundary itself rather than merely post-processing a fixed score (An & Xu, 2025); and PRM calibration matters precisely because an apparently good ranking score may still be a poor decision-grade success estimate (Park et al., 2025). A major open problem is therefore methodological: evaluation should be indexed by the downstream action that confidence governs. The field needs benchmarks and metrics that directly measure decision quality for filtering, stopping, routing, abstaining, covering, and aggregation, rather than assuming that one calibration metric can stand in for all of them.

Challenge 5: Robustness and portability. Confidence utilization methods are typically validated within a fixed model family, prompt format, task distribution, and control protocol. Whether the same signal remains meaningful after any of those ingredients change is much less understood. Verbal confidence is already known to be vulnerable under adversarial attack (Obadinma & Zhu, 2025), and black-box consistency signals depend on how the model reacts to prompt perturbations (Joo et al., 2025). Model-selection systems also inherit a form of portability risk, since their learned or implicit confidence semantics depend on the available model pool and cost-quality frontier (Chen et al., 2023b). At the same time, work such as Superfiltering suggests that some useful control signals can transfer surprisingly well across scale (Li et al., 2024b). The open question is not whether transfer ever occurs, but when. A mature confidence-utilization framework should be able to detect when its assumptions no longer hold under domain shift, prompt shift, model replacement, adversarial manipulation, or protocol change, and then adapt or abstain accordingly.

Taken together, these challenges suggest that the next stage of the field is not simply to design more task-specific confidence heuristics, but to build confidence systems that are semantically interpretable, composable across units and horizons, source-aware, decision-aware, and robust to changing deployment conditions. Progress on these fronts would move confidence utilization beyond a collection of successful tricks and closer to a general theory of confidence as control in LLM systems.

10 Conclusion

This survey argued that confidence in LLM systems is most useful when treated not only as something to be estimated, but as something that changes behavior. Under the unified notation of Section 2, confidence is any reliability-relevant signal attached to a decision unit under a local decision state and then consumed by a downstream policy. Viewed through that lens, confidence-guided control appears across six parallel domains of the LLM lifecycle: training, inference, model selection and cascading, retrieval-augmented generation, risk management, and agentic control. In each domain, confidence matters because it governs actions such as filtering, selection, routing, retrieval, abstention, and search control.

The main lesson of the literature is not that there exists one best confidence signal, but that useful signals differ in semantics, source, unit, and objective. A score that works well for selection may be poor for abstention; a signal that is calibrated in isolation may still be misaligned with routing or retrieval; and a confidence estimate that is meaningful for an answer may not compose cleanly to trajectories, debates, or multi-stage systems. The field has therefore moved beyond confidence estimation alone. The harder problem is to build confidence systems that are interpretable, provenance-aware, composable across units and horizons, robust under shift, and evaluated by the quality of the decisions they support. Progress on these fronts would move the area beyond task-specific heuristics and toward a more general framework for reliable, efficient, and trustworthy control in LLM systems.

11 Limitations

This survey focuses on **confidence utilization** as a control signal rather than confidence estimation or calibration. Readers seeking comprehensive coverage of uncertainty quantification methods should consult complementary surveys.

We primarily survey English-language literature and methods evaluated on English benchmarks; confidence utilization in multilingual and low-resource settings remains comparatively underexplored.

Control policies inherit failure modes of the underlying confidence signal (miscalibration, bias, brittleness under distribution shift or adversarial prompting), which can lead to unsafe acceptance, premature deferral, or unnecessary escalation.

Many methods add latency or compute (sampling, verifiers, multi-stage retrieval), and formal guarantees (e.g., conformal prediction) depend on assumptions and proxies for correctness that may not hold under deployment drift.

Implication. Taken together, these limitations suggest that confidence-as-control should be viewed as a systems design paradigm rather than a drop-in solution: robust deployment requires validating confidence reliability under realistic shift, aligning proxy correctness with application risk, and accounting for compute and human factors alongside model accuracy.

Broader Impact Statement

This survey studies how confidence signals can be used to control the behavior of large language model systems across training, inference, routing, retrieval, risk management, and agentic control. Better confidence utilization could yield meaningful societal benefits, including safer abstention, more reliable escalation to stronger models or human oversight, improved robustness in retrieval and reasoning pipelines, and more trustworthy deployment in high-stakes settings.

At the same time, confidence-guided control introduces its own risks. Confidence signals are heterogeneous, imperfect, and often context-dependent; if they are treated as universally reliable, they may create a false sense of safety or justify incorrect automation decisions. In deployed systems, confidence-based filtering, refusal, routing, or prioritization may also encode hidden biases, suppress minority or out-of-distribution cases, or obscure responsibility when models defer to other components. We therefore view confidence not as a guarantee of correctness, but as a decision instrument whose meaning, robustness, and downstream consequences must be evaluated carefully in context.

Acknowledgments

This research was supported by the National Institute of Standards and Technology under Federal Award ID 60NANB24D231 and by Carnegie Mellon University’s AI Measurement Science and Engineering Center (AIMSEC).

References

- Amro Abbas, Kushal Tirumala, Dániel Simig, Surya Ganguli, and Ari S Morcos. Semdedup: Data-efficient learning at web-scale through semantic deduplication. *arXiv preprint arXiv:2303.09540*, 2023.
- Pranjal Aggarwal, Aman Madaan, Yiming Yang, and Mausam. Let’s sample step by step: Adaptive-consistency for efficient reasoning and coding with llms. In *EMNLP*, 2023.
- Pranjal Aggarwal, Aman Madaan, Ankit Anand, Srividya Pranavi Potharaju, Swaroop Mishra, Pei Zhou, Aditya Gupta, Dheeraj Rajagopal, Karthik Kappaganthu, Yiming Yang, Shyam Upadhyay, Manaal Faruqui, and Mausam. AutoMix: Automatically mixing language models. In *Advances in Neural Information Processing Systems 37 (NeurIPS 2024)*, 2024. doi: 10.52202/079017-4164. URL https://proceedings.neurips.cc/paper_files/paper/2024/hash/ecda225cb187b40ea8edc1f46b03ffda-Abstract-Conference.html.
- Hao An and Yang Xu. Teaching llms to abstain via fine-grained semantic confidence reward. *arXiv preprint arXiv:2510.24020*, 2025.
- Anastasios Nikolas Angelopoulos, Stephen Bates, Adam Fisch, Lihua Lei, and Tal Schuster. Conformal risk control. In *The Twelfth International Conference on Learning Representations*, 2024.
- Akari Asai, Zequi Wu, Yizhong Wang, Avirup Sil, and Hannaneh Hajishirzi. Self-RAG: Learning to retrieve, generate, and critique through self-reflection. In *The Twelfth International Conference on Learning Representations*, 2024. URL <https://openreview.net/forum?id=hSyW5go0v8>. Oral.
- Amos Azaria and Tom Mitchell. The internal state of an llm knows when it’s lying. In *Findings of the Association for Computational Linguistics: EMNLP 2023*, pp. 967–976, 2023.
- Debangshu Banerjee and Aditya Gopalan. Towards reliable alignment: Uncertainty-aware rlhf. *arXiv preprint arXiv:2410.23726*, 2024.
- Amine Barrak, Yosr Fourati, Michael Olchawa, Emna Ksontini, and Khalil Zoghlami. Cargo: A framework for confidence-aware routing of large language models. *arXiv preprint arXiv:2509.14899*, 2025.
- Debashish Chakraborty, Eugene Yang, Daniel Khashabi, Dawn Lawrie, and Kevin Duh. Principled context engineering for rag: Statistical guarantees via conformal prediction. *arXiv preprint arXiv:2511.17908*, 2025.
- Chao Chen, Kai Liu, Ze Chen, Yi Gu, Yue Wu, Mingyuan Tao, Zhihang Fu, and Jieping Ye. INSIDE: LLMs’ internal states retain the power of hallucination detection. In *The Twelfth International Conference on Learning Representations*, 2024a.
- Jiefeng Chen, Jinsung Yoon, Sayna Ebrahimi, Sercan Arik, Tomas Pfister, and Somesh Jha. Adaptation with self-evaluation to improve selective prediction in llms. In *Findings of the Association for Computational Linguistics: EMNLP 2023*, pp. 5190–5213, 2023a.
- Jiuhai Chen and Jonas Mueller. Automated data curation for robust language model fine-tuning. *arXiv preprint arXiv:2403.12776*, 2024.
- Justin Chen, Swarnadeep Saha, and Mohit Bansal. ReConcile: Round-table conference improves reasoning via consensus among diverse LLMs. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pp. 7066–7085, Bangkok, Thailand, 2024b. Association for Computational Linguistics. doi: 10.18653/v1/2024.acl-long.381. URL <https://aclanthology.org/2024.acl-long.381/>.
- Lingjiao Chen, Matei Zaharia, and James Zou. Frugalgpt: How to use large language models while reducing cost and improving performance. *arXiv preprint arXiv:2305.05176*, 2023b.
- Wang Chen, Guanqiang Qi, Weikang Li, Yang Li, Deguo Xia, and Jizhou Huang. Pairs: Parametric-verified adaptive information retrieval and selection for efficient rag. *arXiv preprint arXiv:2508.04057*, 2025a.

-
- Xinyun Chen, Renat Aksitov, Uri Alon, Jie Ren, Kefan Xiao, Pengcheng Yin, Sushant Prakash, Charles Sutton, Xuezhi Wang, and Denny Zhou. Universal self-consistency for large language model generation. *arXiv preprint arXiv:2311.17311*, 2023c.
- Zhuokun Chen, Zeren Chen, Jiahao He, Lu Sheng, Minghui Tan, Jianfei Cai, and Bohan Zhuang. R-stitch: Dynamic trajectory stitching for efficient reasoning. *arXiv preprint arXiv:2507.17307*, 2025b. doi: 10.48550/arXiv.2507.17307. URL <https://arxiv.org/abs/2507.17307>.
- John Cherian, Isaac Gibbs, and Emmanuel Candes. Large language model validity via enhanced conformal prediction methods. *Advances in Neural Information Processing Systems*, 37:114812–114842, 2024.
- Yu-Neng Chuang, Prathusha Kameswara Sarma, Parikshit Gopalan, John Boccio, Sara Bolouki, Xia Hu, and Helen Zhou. Learning to route LLMs with confidence tokens. In *Proceedings of the 42nd International Conference on Machine Learning*, volume 267 of *Proceedings of Machine Learning Research*, pp. 10859–10878. PMLR, 2025a. URL <https://proceedings.mlr.press/v267/chuang25b.html>.
- Yu-Neng Chuang, Leisheng Yu, Guanchu Wang, Lizhe Zhang, Zirui Liu, Xuanning Cai, Yang Sui, Vladimir Braverman, and Xia Hu. Confident or seek stronger: Exploring uncertainty-based on-device LLM routing from benchmarking to generalization. *arXiv preprint arXiv:2502.04428*, 2025b. URL <https://arxiv.org/abs/2502.04428>.
- Yung-Sung Chuang, Yujia Xie, Hongyin Luo, Yoon Kim, James R. Glass, and Pengcheng He. DoLa: Decoding by contrasting layers improves factuality in large language models. In *The Twelfth International Conference on Learning Representations*, 2024. URL <https://openreview.net/forum?id=Th6NyL07na>. Poster.
- Jeremy Cole, Michael Zhang, Dan Gillick, Julian Eisenschlos, Bhuwan Dhingra, and Jacob Eisenstein. Selectively answering ambiguous questions. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pp. 530–543, 2023.
- Yu Cui, Hang Fu, Haibin Zhang, Licheng Wang, and Cong Zuo. Free-mad: Consensus-free multi-agent debate. *arXiv preprint arXiv:2509.11035*, 2025. URL <https://arxiv.org/abs/2509.11035>.
- Jasper Dekoninck, Maximilian Baader, and Martin Vechev. A unified approach to routing and cascading for llms. *arXiv preprint arXiv:2410.10347*, 2024.
- Dujian Ding, Ankur Mallick, Chi Wang, Robert Sim, Subhabrata Mukherjee, Victor Ruhle, Laks VS Lakshmanan, and Ahmed Hassan Awadallah. Hybrid LLM: Cost-efficient and quality-aware query routing. In *ICLR*, 2024.
- He Du, Bowen Li, Chengxing Xie, Chang Gao, Kai Chen, and Dacheng Tao. Confidence as a reward: Transforming llms into reward models. *arXiv preprint arXiv:2510.13501*, 2025.
- Ekaterina Fadeeva, Aleksandr Rubashevskii, Roman Vashurin, Shehzaad Dhuliawala, Artem Shelmanov, Timothy Baldwin, Preslav Nakov, Mrinmaya Sachan, and Maxim Panov. Faithfulness-aware uncertainty quantification for fact-checking the output of retrieval augmented generation. *arXiv preprint arXiv:2505.21072*, 2025.
- Wei Fan, JinYi Yoon, and Bo Ji. imad: Intelligent multi-agent debate for efficient and accurate llm inference. *arXiv preprint arXiv:2511.11306*, 2025. URL <https://arxiv.org/abs/2511.11306>.
- Sebastian Farquhar, Jannik Kossen, Lorenz Kuhn, and Yarin Gal. Detecting hallucinations in large language models using semantic entropy. *Nature*, 630(8017):625–630, 2024.
- Naihe Feng, Yi Sui, Shiyi Hou, Jesse C Cresswell, and Ga Wu. Response quality assessment for retrieval-augmented generation via conditional conformal factuality. In *Proceedings of the 48th International ACM SIGIR Conference on Research and Development in Information Retrieval*, pp. 2832–2836, 2025.
- Renyu Fu and Guibo Luo. SeLaR: Selective latent reasoning in large language models. *arXiv preprint arXiv:2604.08299*, 2026. doi: 10.48550/arXiv.2604.08299. URL <https://arxiv.org/abs/2604.08299>. Camera-ready for ACL 2026.

-
- Yichao Fu, Xuwei Wang, Yuandong Tian, and Jiawei Zhao. Deep think with confidence. *arXiv preprint arXiv:2508.15260*, 2025.
- Jiahui Geng, Fengyu Cai, Yuxia Wang, Heinz Koepl, Preslav Nakov, and Iryna Gurevych. A survey of confidence estimation and calibration in large language models. In *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, pp. 6577–6595, 2024.
- Saumya Goswami and Siddharth Kurra. Halt-rag: A task-adaptable framework for hallucination detection with calibrated nli ensembles and abstention. *arXiv preprint arXiv:2509.07475*, 2025.
- Tobias Groot and Matias Valdenegro Toro. Overconfidence is key: Verbalized uncertainty evaluation in large language and vision-language models. In Anaelia Ovalle, Kai-Wei Chang, Yang Trista Cao, Ninareh Mehrabi, Jieyu Zhao, Aram Galstyan, Jwala Dhamala, Anoop Kumar, and Rahul Gupta (eds.), *Proceedings of the 4th Workshop on Trustworthy Natural Language Processing (TrustNLP 2024)*, pp. 145–171, Mexico City, Mexico, June 2024. Association for Computational Linguistics.
- Yu Gui, Ying Jin, and Zhimei Ren. Conformal alignment: Knowing when to trust foundation models with guarantees. *Advances in Neural Information Processing Systems*, 37:73884–73919, 2024. URL <https://openreview.net/forum?id=YzyCEJlV9Z>. NeurIPS 2024 poster.
- Neha Gupta, Harikrishna Narasimhan, Wittawat Jitkrittum, Ankit Singh Rawat, Aditya Krishna Menon, and Sanjiv Kumar. Language model cascades: Token-level uncertainty and beyond. *arXiv preprint arXiv:2404.10136*, 2024.
- Jiatong Han, Jannik Kossen, Muhammed Razzak, Lisa Schut, Shreshth A Malik, and Yarin Gal. Semantic entropy probes: Robust and cheap hallucination detection in LLMs. In *ICML 2024 Workshop on Foundation Models in the Wild*, 2024.
- Jindong Han, Hao Liu, Jun Fang, Naiqiang Tan, and Hui Xiong. Automatic instruction data selection for large language models via uncertainty-aware influence maximization. In *Proceedings of the ACM on Web Conference 2025*, pp. 4969–4979, 2025.
- Cheng Peng Huang and Hao-Yuan Chen. Delta–contrastive decoding mitigates text hallucinations in large language models. *arXiv preprint arXiv:2502.05825*, 2025.
- Chengsong Huang, Langlin Huang, Jixuan Leng, Jiacheng Liu, and Jiaxin Huang. Efficient test-time scaling via self-calibration. *arXiv preprint arXiv:2503.00031*, 2025a.
- Haiduo Huang, Jiangcheng Song, Yadong Zhang, and Pengju Ren. Selectkd: Selective token-weighted knowledge distillation for llms. *arXiv preprint arXiv:2510.24021*, 2025b.
- Lei Huang, Weijiang Yu, Weitao Ma, Weihong Zhong, Zhangyin Feng, Haotian Wang, Qianglong Chen, Weihua Peng, Xiaocheng Feng, Bing Qin, and Ting Liu. A survey on hallucination in large language models: Principles, taxonomy, challenges, and open questions. *ACM Transactions on Information Systems*, 43(2): 42:1–42:55, 2025c. doi: 10.1145/3703155. URL <https://doi.org/10.1145/3703155>.
- Zhiqi Huang, Vivek Datla, Chenyang Zhu, Alf Samuel, Daben Liu, Anoop Kumar, and Ritesh Soni. Confidence-based response abstinence: Improving llm trustworthiness via activation-based uncertainty estimation. In *Proceedings of the 2nd Workshop on Uncertainty-Aware NLP (UncertaiNLP 2025)*, pp. 184–193, 2025d.
- Tomoaki Isoda. Skill-rag: Self-knowledge induced learning and filtering for retrieval-augmented generation. *arXiv preprint arXiv:2509.20377*, 2025.
- Eunhye Jeong and Yong Suk Choi. Acr: Adaptive confidence re-scoring for reliable answer selection among multiple candidates. *Applied Sciences*, 15(17):9587, 2025.

-
- Soyeong Jeong, Jinheon Baek, Sukmin Cho, Sung Ju Hwang, and Jong C Park. Adaptive-rag: Learning to adapt retrieval-augmented large language models through question complexity. *arXiv preprint arXiv:2403.14403*, 2024.
- Ziwei Ji, Lei Yu, Yeskendir Koishikenov, Yejin Bang, Anthony Hartshorn, Alan Schelten, Cheng Zhang, Pascale Fung, and Nicola Cancedda. Calibrating verbal uncertainty as a linear feature to reduce hallucinations. *arXiv preprint arXiv:2503.14477*, 2025.
- Zhengbao Jiang, Frank F Xu, Luyu Gao, Zhiqing Sun, Qian Liu, Jane Dwivedi-Yu, Yiming Yang, Jamie Callan, and Graham Neubig. Active retrieval augmented generation. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pp. 7969–7992, 2023.
- Zhengping Jiang, Anqi Liu, and Benjamin Van Durme. Conformal linguistic calibration: Trading-off between factuality and specificity. *arXiv preprint arXiv:2502.19110*, 2025. doi: 10.48550/arXiv.2502.19110. URL <https://arxiv.org/abs/2502.19110>.
- Haoxiang Jin, Ronghan Li, Qiguang Miao, and Zixiang Lu. Rethinking LLM parametric knowledge as post-retrieval confidence for dynamic retrieval and reranking. *arXiv preprint arXiv:2509.06472*, 2025. URL <https://arxiv.org/abs/2509.06472>.
- Seongho Joo, Kyungmin Min, Jahyun Koo, and Kyomin Jung. Black-box hallucination detection via consistency under the uncertain expression. *arXiv preprint arXiv:2509.21999*, 2025.
- Saurav Kadavath, Tom Conerly, Amanda Askell, Tom Henighan, Dawn Drain, Ethan Perez, Nicholas Schiefer, Zac Hatfield-Dodds, Nova DasSarma, Eli Tran-Johnson, Scott Johnston, Sheer El-Showk, Andy Jones, Nelson Elhage, Tristan Hume, Anna Chen, Yuntao Bai, Sam Bowman, Stanislav Fort, Deep Ganguli, Danny Hernandez, Josh Jacobson, Jackson Kernion, Shauna Kravec, Liane Lovitt, Kamal Ndousse, Catherine Olsson, Sam Ringer, Dario Amodei, Tom Brown, Jack Clark, Nicholas Joseph, Ben Mann, Sam McCandlish, Chris Olah, and Jared Kaplan. Language models (mostly) know what they know. *arXiv preprint arXiv:2207.05221*, 2022. URL <https://arxiv.org/abs/2207.05221>.
- Lars Benedikt Kaesberg, Jonas Becker, Jan Philip Wahle, Terry Ruas, and Bela Gipp. Voting or consensus? decision-making in multi-agent debate. In *Findings of the Association for Computational Linguistics: ACL 2025*, pp. 11640–11671, Vienna, Austria, 2025. Association for Computational Linguistics. doi: 10.18653/v1/2025.findings-acl.606. URL <https://aclanthology.org/2025.findings-acl.606/>.
- Anant Khandelwal, Manish Gupta, and Puneet Agrawal. Cocoa: Confidence-and context-aware adaptive decoding for resolving knowledge conflicts in large language models. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, pp. 6846–6866, 2025.
- Joongwon Kim, Anirudh Goyal, Liang Tan, Hannaneh Hajishirzi, Srinivasan Iyer, and Tianlu Wang. AS-TRO: Teaching language models to reason by reflecting and backtracking in-context. *arXiv preprint arXiv:2507.00417*, 2025. URL <https://arxiv.org/abs/2507.00417>.
- Ranganath Krishnan, Piyush Khanna, and Omesh Tickoo. Enhancing trust in large language models with uncertainty-aware fine-tuning. *arXiv preprint arXiv:2412.02904*, 2024.
- Bhawesh Kumar, Charlie Lu, Gauri Gupta, Anil Palepu, David Bellamy, Ramesh Raskar, and Andrew Beam. Conformal prediction with large language models for multi-choice question answering. *arXiv preprint arXiv:2305.18404*, 2023.
- Hyunseok Lee, Seunghyuk Oh, Jaehyung Kim, Jinwoo Shin, and Jihoon Tack. ReVISE: Learning to refine at test-time via intrinsic self-verification. In *Proceedings of the 42nd International Conference on Machine Learning*, volume 267 of *Proceedings of Machine Learning Research*, pp. 33616–33634. PMLR, 2025. URL <https://proceedings.mlr.press/v267/lee25ab.html>.
- Jixuan Leng, Chengsong Huang, Banghua Zhu, and Jiaxin Huang. Taming overconfidence in llms: Reward calibration in rlhf. *arXiv preprint arXiv:2410.09724*, 2024.

-
- Yaniv Leviathan, Matan Kalman, and Yossi Matias. Fast inference from transformers via speculative decoding. In *ICML*, 2023.
- Bo Li, Tian Tian, Zhenghua Xu, Hao Cheng, Shikun Zhang, and Wei Ye. Modeling uncertainty trends for timely retrieval in dynamic rag. *arXiv preprint arXiv:2511.09980*, 2025a.
- Haodong Li, Jingqi Zhang, Xiao Cheng, Peihua Mai, Haoyu Wang, and Yan Pang. As if we’ve met before: Lms exhibit certainty in recognizing seen files. *arXiv preprint arXiv:2511.15192*, 2025b.
- Jiaqi Li, Yixuan Tang, and Yi Yang. Know the unknown: An uncertainty-sensitive method for llm instruction tuning. In *Findings of the Association for Computational Linguistics: ACL 2025*, pp. 2972–2989, 2025c.
- Ming Li, Lichang Chen, Jiuhai Chen, Shwai He, Jiuxiang Gu, and Tianyi Zhou. Selective reflection-tuning: Student-selected data recycling for llm instruction-tuning. In *Findings of the Association for Computational Linguistics: ACL 2024*, pp. 16189–16211, Bangkok, Thailand, 2024a. Association for Computational Linguistics. doi: 10.18653/v1/2024.findings-acl.958. URL <https://aclanthology.org/2024.findings-acl.958/>.
- Ming Li, Yong Zhang, Shwai He, Zhitao Li, Hongyu Zhao, Jianzong Wang, Ning Cheng, and Tianyi Zhou. Superfiltering: Weak-to-strong data filtering for fast instruction-tuning. *arXiv preprint arXiv:2402.00530*, 2024b.
- Ming Li, Yong Zhang, Zhitao Li, Jiuhai Chen, Lichang Chen, Ning Cheng, Jianzong Wang, Tianyi Zhou, and Jing Xiao. From quantity to quality: Boosting llm performance with self-guided data selection for instruction tuning. In *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, pp. 7602–7635, 2024c.
- Pengyi Li, Matvey Skripkin, Alexander Zubrey, Andrey Kuznetsov, and Ivan Oseledets. Confidence is all you need: Few-shot rl fine-tuning of language models. *arXiv preprint arXiv:2506.06395*, 2025d.
- Shuo Li, Sangdon Park, Insup Lee, and Osbert Bastani. Traq: Trustworthy retrieval augmented question answering via conformal prediction. In *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, pp. 3799–3821, 2024d.
- Xiang Lisa Li, Ari Holtzman, Daniel Fried, Percy Liang, Jason Eisner, Tatsunori B Hashimoto, Luke Zettlemoyer, and Mike Lewis. Contrastive decoding: Open-ended text generation as optimization. In *Proceedings of the 61st annual meeting of the association for computational linguistics (volume 1: Long papers)*, pp. 12286–12312, 2023.
- Yibo Li, Miao Xiong, Jiaying Wu, and Bryan Hooi. Conftuner: Training large language models to express their confidence verbally. In *The Thirty-ninth Annual Conference on Neural Information Processing Systems*, 2025e.
- Yubo Li, Yidi Miao, Xueying Ding, Ramayya Krishnan, and Rema Padman. Firm or fickle? evaluating large language models consistency in sequential interactions. In *Findings of the Association for Computational Linguistics: ACL 2025*, pp. 6679–6700, Vienna, Austria, July 2025f. Association for Computational Linguistics. ISBN 979-8-89176-256-5.
- Zixuan Li, Jing Xiong, Fanghua Ye, Chuanyang Zheng, Xun Wu, Jianqiao Lu, Zhongwei Wan, Xiaodan Liang, Chengming Li, Zhenan Sun, Lingpeng Kong, and Ngai Wong. UncertaintyRAG: Span-level uncertainty enhanced long-context modeling for retrieval-augmented generation. *arXiv preprint arXiv:2410.02719*, 2024e. URL <https://arxiv.org/abs/2410.02719>.
- Hunter Lightman, Vineet Kosaraju, Yura Burda, Harri Edwards, Bowen Baker, Teddy Lee, Jan Leike, John Schulman, Ilya Sutskever, and Karl Cobbe. Let’s verify step by step. In *ICLR*, 2024.

-
- Zijie Lin and Bryan Hooi. Enhancing multi-agent debate system performance via confidence expression. In *Findings of the Association for Computational Linguistics: EMNLP 2025*, pp. 6453–6471, Suzhou, China, 2025. Association for Computational Linguistics. doi: 10.18653/v1/2025.findings-emnlp.343. URL <https://aclanthology.org/2025.findings-emnlp.343/>.
- Aiwei Liu, Haoping Bai, Zhiyun Lu, Yanchao Sun, Xiang Kong, Xiaoming Simon Wang, Jiulong Shan, Albin Madappally Jose, Xiaojiang Liu, Lijie Wen, Philip S. Yu, and Meng Cao. TIS-DPO: Token-level importance sampling for direct preference optimization with estimated weights. In *The Thirteenth International Conference on Learning Representations*, 2025a. URL <https://openreview.net/forum?id=oF6e2WwxX0>. Poster.
- Haotian Liu, Shuo Wang, and Hongteng Xu. c^2 gspg: Confidence-calibrated group sequence policy gradient towards self-aware reasoning. *arXiv preprint arXiv:2509.23129*, 2025b.
- Liangxin Liu, Xuebo Liu, Derek F Wong, Dongfang Li, Ziyi Wang, Baotian Hu, and Min Zhang. Selectit: Selective instruction tuning for large language models via uncertainty-aware self-reflection. *arXiv preprint arXiv:2402.16705*, 2024a.
- Xiaoou Liu, Tiejun Chen, Longchao Da, Chacha Chen, Zhen Lin, and Hua Wei. Uncertainty quantification and confidence calibration in large language models: A survey. *arXiv preprint arXiv:2503.15850*, 2025c.
- Yueyue Liu, Hongyu Zhang, Yuantian Miao, Van-Hoang Le, and Zhiqiang Li. Optllm: Optimal assignment of queries to large language models. In *2024 IEEE International Conference on Web Services (ICWS)*, pp. 788–798. IEEE, 2024b.
- Junjie Lu, Yuliang Liu, Chaofeng Qu, Wei Shen, Zhouhan Lin, and Min Xu. Enhancing llm reasoning via non-human-like reasoning path preference optimization. *arXiv preprint arXiv:2510.11104*, 2025.
- Panagiotis Lymperopoulos and Vasanth Sarathy. Tools in the loop: Quantifying uncertainty of llm question answering systems that use tools. *arXiv preprint arXiv:2505.16113*, 2025.
- Aman Madaan, Niket Tandon, Prakhar Gupta, Skyler Hallinan, Luyu Gao, Sarah Wiegrefe, Uri Alon, Nouha Dziri, Shrimai Prabhumoye, Yiming Yang, Shashank Gupta, Bodhisattwa Prasad Majumder, Katherine Hermann, Sean Welleck, Amir Yazdanbakhsh, and Peter Clark. Self-refine: Iterative refinement with self-feedback. *Advances in Neural Information Processing Systems*, 36:46534–46594, 2023. URL https://proceedings.neurips.cc/paper_files/paper/2023/hash/91eddf07232fb1b55a505a9e9f6c0ff3-Abstract-Conference.html.
- Alex Mallen, Akari Asai, Victor Zhong, Rajarshi Das, Daniel Khoshdel, and Hannaneh Hajishirzi. When not to trust language models: Investigating effectiveness of parametric and non-parametric memories. In *ACL*, 2023.
- Potsawee Manakul, Adian Liusie, and Mark Gales. Selfcheckgpt: Zero-resource black-box hallucination detection for generative large language models. In *Proceedings of the 2023 conference on empirical methods in natural language processing*, pp. 9004–9017, 2023.
- Yu Meng, Mengzhou Xia, and Danqi Chen. SimPO: Simple preference optimization with a reference-free reward. In *NeurIPS*, 2024.
- Sewon Min, Kalpesh Krishna, Xinxin Lyu, Mike Lewis, Wen-tau Yih, Pang Wei Koh, Mohit Iyyer, Luke Zettlemoyer, and Hannaneh Hajishirzi. FActScore: Fine-grained atomic evaluation of factual precision in long form text generation. In *EMNLP*, 2023.
- Christopher Mohri and Tatsunori Hashimoto. Language models with conformal factuality guarantees. In *International Conference on Machine Learning*, pp. 36029–36047. PMLR, 2024.
- Niklas Muennighoff, Zitong Yang, Weijia Shi, Xiang Lisa Li, Li Fei-Fei, Hannaneh Hajishirzi, Luke Zettlemoyer, Percy Liang, Emmanuel Candès, and Tatsunori Hashimoto. s1: Simple test-time scaling. *arXiv preprint arXiv:2501.19393*, 2025.

-
- Diyana Muhammed, Gollam Rabby, and Sören Auer. Selfcheckagent: Zero-resource hallucination detection in generative large language models. *arXiv preprint arXiv:2502.01812*, 2025.
- William Muldrew, Peter Hayes, Mingtian Zhang, and David Barber. Active preference learning for large language models. *arXiv preprint arXiv:2402.08114*, 2024.
- Harikrishna Narasimhan, Wittawat Jitkrittum, Ankit Singh Rawat, Seungyeon Kim, Neha Gupta, Aditya Krishna Menon, and Sanjiv Kumar. Faster cascades via speculative decoding. *arXiv preprint arXiv:2405.19261*, 2024.
- Cheng Niu, Yuanhao Wu, Juno Zhu, Siliang Xu, KaShun Shum, Randy Zhong, Juntong Song, and Tong Zhang. RAGTruth: A hallucination corpus for developing trustworthy retrieval-augmented language models. In *ACL*, 2024.
- Stephen Obadinma and Xiaodan Zhu. On the robustness of verbal confidence of llms in adversarial attacks. *arXiv preprint arXiv:2507.06489*, 2025.
- Stephen Obadinma, Xiaodan Zhu, and Hongyu Guo. Calibration attacks: A comprehensive study of adversarial attacks on model confidence. *arXiv preprint arXiv:2401.02718*, 2024.
- Isaac Ong, Amjad Almahairi, Vincent Wu, Wei-Lin Chiang, Tianhao Wu, Joseph E. Gonzalez, M. Waleed Kadous, and Ion Stoica. RouteLLM: Learning to route LLMs with preference data. *arXiv preprint arXiv:2406.18665*, 2024. doi: 10.48550/arXiv.2406.18665. URL <https://arxiv.org/abs/2406.18665>.
- Yijun Pan, Taiwei Shi, Jieyu Zhao, and Jiaqi W Ma. Detecting and filtering unsafe training data via data attribution. *arXiv preprint arXiv:2502.11411*, 2025.
- Jinlong Pang, Na Di, Zhaowei Zhu, Jiaheng Wei, Hao Cheng, Chen Qian, and Yang Liu. Token cleaning: Fine-grained data selection for llm supervised fine-tuning. *arXiv preprint arXiv:2502.01968*, 2025.
- Young-Jin Park, Kristjan Greenewald, Kaveh Alim, Hao Wang, and Navid Azizan. Know what you don’t know: Uncertainty calibration of process reward models. In *The Thirty-ninth Annual Conference on Neural Information Processing Systems*, 2025.
- Hao Peng, Yunjia Qi, Xiaozhi Wang, Zijun Yao, Bin Xu, Lei Hou, and Juanzi Li. Agentic reward modeling: Integrating human preferences with verifiable correctness signals for reliable reward systems. *arXiv preprint arXiv:2502.19328*, 2025. URL <https://arxiv.org/abs/2502.19328>.
- Rhitabrat Pokharel, Yufei Tao, and Ameeta Agrawal. Capo: Confidence aware preference optimization learning for multilingual preferences. *arXiv preprint arXiv:2511.07691*, 2025.
- Mihir Prabhudesai, Lili Chen, Alex Ippoliti, Katerina Fragkiadaki, Hao Liu, and Deepak Pathak. Maximizing confidence alone improves reasoning. *arXiv preprint arXiv:2505.22660*, 2025.
- Ziqing Qiao, Yongheng Deng, Jiali Zeng, Dong Wang, Lai Wei, Guanbo Wang, Fandong Meng, Jie Zhou, Ju Ren, and Yaoxue Zhang. Concise: Confidence-guided compression in step-by-step efficient reasoning. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, pp. 8021–8040, 2025.
- Victor Quach, Adam Fisch, Tal Schuster, Adam Yala, Jae Ho Sohn, Tommi S. Jaakkola, and Regina Barzilay. Conformal language modeling. In *The Twelfth International Conference on Learning Representations*, 2024.
- Stephan Rabanser, Nathalie Rauschmayr, Achin Kulshrestha, Petra Poklukar, Wittawat Jitkrittum, Sean Augenstein, Congchao Wang, and Federico Tombari. Gatekeeper: Improving model cascades through confidence tuning. In *The Thirty-ninth Annual Conference on Neural Information Processing Systems*, 2025.
- Rafael Rafailov, Archit Sharma, Eric Mitchell, Christopher D Manning, Stefano Ermon, and Chelsea Finn. Direct preference optimization: Your language model is secretly a reward model. *Advances in neural information processing systems*, 36:53728–53741, 2023.

-
- Amir Hossein Rahmati, Sanket Jantre, Weifeng Zhang, Yucheng Wang, Byung-Jun Yoon, Nathan M Urban, and Xiaoning Qian. C-lora: Contextual low-rank adaptation for uncertainty estimation in large language models. *arXiv preprint arXiv:2505.17773*, 2025.
- Vipula Rawte, Amit Sheth, and Amitava Das. A survey of hallucination in large foundation models. *arXiv preprint arXiv:2309.05922*, 2023.
- Jie Ren, Yao Zhao, Tu Vu, Peter J. Liu, and Balaji Lakshminarayanan. Self-evaluation improves selective generation in large language models. In Javier Antorán, Arno Blaas, Kelly Buchanan, Fan Feng, Vincent Fortuin, Sahra Ghalebikesabi, Andreas Kriegler, Ian Mason, David Rohde, Francisco J. R. Ruiz, Tobias Uelwer, Yubin Xie, and Rui Yang (eds.), *Proceedings on "I Can't Believe It's Not Better: Failure Modes in the Age of Foundation Models" at NeurIPS 2023 Workshops*, volume 239 of *Proceedings of Machine Learning Research*, pp. 49–64. PMLR, 16 Dec 2023.
- Pouria Rouzrokh, Shahriar Faghani, Cooper U Gamble, Moein Shariatnia, and Bradley J Erickson. Conflare: conformal large language model retrieval. *arXiv preprint arXiv:2404.04287*, 2024.
- Maxon Rubin-Toles, Maya Gambhir, Keshav Ramji, Aaron Roth, and Surbhi Goel. Conformal language model reasoning with coherent factuality. In *The Thirteenth International Conference on Learning Representations*, 2025.
- Noveen Sachdeva, Benjamin Coleman, Wang-Cheng Kang, Jianmo Ni, Lichan Hong, Ed H Chi, James Caverlee, Julian McAuley, and Derek Zhiyuan Cheng. How to train data-efficient llms. *arXiv preprint arXiv:2402.09668*, 2024.
- Soham Shah and Kumar Shridhar. Select-then-route : Taxonomy guided routing for LLMs. In Saloni Potdar, Lina Rojas-Barahona, and Sebastien Montella (eds.), *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing: Industry Track*, pp. 425–441, Suzhou (China), November 2025. Association for Computational Linguistics. ISBN 979-8-89176-333-3.
- Haizhou Shi, Ye Liu, Bo Pang, Zeyu Leo Liu, Hao Wang, Silvio Savarese, Caiming Xiong, Yingbo Zhou, and Semih Yavuz. SSR: Socratic self-refine for large language model reasoning. *arXiv preprint arXiv:2511.10621*, 2025. URL <https://arxiv.org/abs/2511.10621>.
- Noah Shinn, Federico Cassano, Ashwin Gopinath, Karthik R. Narasimhan, and Shunyu Yao. Reflexion: Language agents with verbal reinforcement learning. *Advances in Neural Information Processing Systems*, 36, 2023. URL <https://openreview.net/forum?id=vAE1hFcKW6>. Poster.
- Ola Shorinwa, Zhiting Mei, Justin Lidard, Allen Z Ren, and Anirudha Majumdar. A survey on uncertainty quantification of large language models: Taxonomy, open research challenges, and future directions. *ACM Computing Surveys*, 2025.
- Charlie Victor Snell, Jaehoon Lee, Kelvin Xu, and Aviral Kumar. Scaling LLM test-time compute optimally can be more effective than scaling parameters for reasoning. In *The Thirteenth International Conference on Learning Representations*, 2025. URL <https://openreview.net/forum?id=4FWAwZtd2n>. Oral.
- Zeen Song, Zihao Ma, Wenwen Qiang, Changwen Zheng, and Gang Hua. Adaptive uncertainty-aware tree search for robust reasoning. *arXiv preprint arXiv:2602.06493*, 2026.
- Heydar Soudani, Evangelos Kanoulas, and Faegheh Hasibi. Why uncertainty estimation methods fall short in rag: An axiomatic analysis. *arXiv preprint arXiv:2505.07459*, 2025.
- Jiayuan Su, Jing Luo, Hongwei Wang, and Lu Cheng. Api is enough: Conformal prediction for large language models without logit-access. In *Findings of the Association for Computational Linguistics: EMNLP 2024*, pp. 979–995, 2024a.
- Weihang Su, Yichen Tang, Qingyao Ai, Zhijing Wu, and Yiqun Liu. Dragin: dynamic retrieval augmented generation based on the information needs of large language models. *arXiv preprint arXiv:2403.10081*, 2024b.

-
- Xin Sun, Jianan Xie, Zhongqi Chen, Qiang Liu, Shu Wu, Yuehe Chen, Bowen Song, Zilei Wang, Weiqiang Wang, and Liang Wang. Divide-then-align: Honest alignment based on the knowledge boundary of rag. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pp. 11461–11480, 2025.
- Zhongxiang Sun, Xiaoxue Zang, Kai Zheng, Yang Song, Jun Xu, Xiao Zhang, Weijie Yu, and Han Li. Redeeep: Detecting hallucination in retrieval-augmented generation via mechanistic interpretability. *arXiv preprint arXiv:2410.11414*, 2024.
- Shuchang Tao, Liuyi Yao, Hanxing Ding, Yuexiang Xie, Qi Cao, Fei Sun, Jinyang Gao, Huawei Shen, and Bolin Ding. When to trust llms: Aligning confidence with response quality. *arXiv preprint arXiv:2404.17287*, 2024.
- Amir Taubenfeld, Tom Sheffer, Eran Ofek, Amir Feder, Ariel Goldstein, Zorik Gekhman, and Gal Yona. Confidence improves self-consistency in LLMs. In *Findings of the Association for Computational Linguistics: ACL 2025*, pp. 20090–20111, Vienna, Austria, 2025. Association for Computational Linguistics. doi: 10.18653/v1/2025.findings-acl.1030. URL <https://aclanthology.org/2025.findings-acl.1030/>.
- Katherine Tian, Eric Mitchell, Allan Zhou, Archit Sharma, Rafael Rafailov, Huaxiu Yao, Chelsea Finn, and Christopher D Manning. Just ask for calibration: Strategies for eliciting calibrated confidence scores from language models fine-tuned with human feedback. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pp. 5433–5442, 2023.
- Zailong Tian, Zhuoheng Han, Yanzhe Chen, Haozhe Xu, Xi Yang, Richeng Xuan, Houfeng Wang, and Lizi Liao. Overconfidence in llm-as-a-judge: Diagnosis and confidence-driven solution. *arXiv preprint arXiv:2508.06225*, 2025.
- Benedict Aaron Tjandra, Muhammed Razzak, Jannik Kossen, Kunal Handa, and Yarin Gal. Fine-tuning large language models to appropriately abstain with semantic entropy. In *Neurips Safe Generative AI Workshop 2024*, 2024.
- SM Tonmoy, SM Zaman, Vinija Jain, Anku Rani, Vipula Rawte, Aman Chadha, and Amitava Das. A comprehensive survey of hallucination mitigation techniques in large language models. *arXiv preprint arXiv:2401.01313*, 2024.
- Sahil Tripathi, Md Tabrez Nafis, Imran Hussain, and Jiechao Gao. The confidence paradox: Can llm know when it’s wrong. *arXiv preprint arXiv:2506.23464*, 2025.
- Neeraj Varshney and Chitta Baral. Model cascading: Towards jointly improving efficiency and accuracy of NLP systems. In *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing*, pp. 11007–11021, Abu Dhabi, United Arab Emirates, 2022. Association for Computational Linguistics. doi: 10.18653/v1/2022.emnlp-main.756. URL <https://aclanthology.org/2022.emnlp-main.756/>.
- Junlin Wang, Jue Wang, Ben Athiwaratkun, Ce Zhang, and James Zou. Mixture-of-agents enhances large language model capabilities. In *ICLR*, 2025a.
- Peiyi Wang, Lei Li, Zhihong Shao, Runxin Xu, Damai Dai, Yifei Li, Deli Chen, Yu Wu, and Zhifang Sui. Math-shepherd: Verify and reinforce LLMs step-by-step without human annotations. In *ACL*, 2024.
- Xuezhi Wang, Jason Wei, Dale Schuurmans, Quoc V. Le, Ed H. Chi, Sharan Narang, Aakanksha Chowdhery, and Denny Zhou. Self-consistency improves chain of thought reasoning in language models. In *The Eleventh International Conference on Learning Representations*, 2023a. URL <https://openreview.net/forum?id=1PL1NIMMrw>. Poster.
- Yile Wang, Peng Li, Maosong Sun, and Yang Liu. Self-knowledge guided retrieval augmentation for large language models. *arXiv preprint arXiv:2310.05002*, 2023b.
- Zhiruo Wang, Jun Araki, Zhengbao Jiang, Md Rizwan Parvez, and Graham Neubig. Learning to filter context for retrieval-augmented generation. *arXiv preprint arXiv:2311.08377*, 2023c.

-
- Zihan Wang, Zihan Liang, Zhou Shao, Yufei Ma, Huangyu Dai, Ben Chen, Lingtao Mao, Chenyi Lei, Yuqing Ding, and Han Li. Infogain-rag: Boosting retrieval-augmented generation through document information gain-based reranking and filtering. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, pp. 7201–7215, 2025b.
- Chengwei Wei, Kee Kiat Koo, Amir Tavanaei, and Karim Bouyarmane. Confidence-aware sub-structure beam search (cabs): Mitigating hallucination in structured data generation with large language models. *arXiv preprint arXiv:2406.00069*, 2024.
- Bingbing Wen, Jihan Yao, Shangbin Feng, Chenjun Xu, Yulia Tsvetkov, Bill Howe, and Lucy Lu Wang. Know your limits: A survey of abstention in large language models. *Transactions of the Association for Computational Linguistics*, 13:529–556, 2025.
- Haolun Wu, Zhenkun Li, and Lingyao Li. Can llm agents really debate? a controlled study of multi-agent debate in logical reasoning. *arXiv preprint arXiv:2511.07784*, 2025a. URL <https://arxiv.org/abs/2511.07784>.
- Jiayun Wu, Jiashuo Liu, Zhiyuan Zeng, Tianyang Zhan, and Wenhao Huang. Mitigating llm hallucination via behaviorally calibrated reinforcement learning. *arXiv preprint arXiv:2512.19920*, 2025b.
- Junkang Wu, Yuexiang Xie, Zhengyi Yang, Jiancan Wu, Jinyang Gao, Bolin Ding, Xiang Wang, and Xiangnan He. β -DPO: Direct preference optimization with dynamic β . *Advances in Neural Information Processing Systems*, 37, 2024. doi: 10.52202/079017-4128. URL https://proceedings.neurips.cc/paper_files/paper/2024/hash/ea888178abdb6fc233226d12321d754f-Abstract-Conference.html.
- Qinzhao Wu, Pengzhi Gao, Wei Liu, and Jian Luan. BacktrackAgent: Enhancing GUI agent with error detection and backtracking mechanism. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, pp. 4250–4272, Suzhou, China, 2025c. Association for Computational Linguistics. doi: 10.18653/v1/2025.emnlp-main.212. URL <https://aclanthology.org/2025.emnlp-main.212/>.
- HuaJun Xi, Jianguo Huang, Kangdao Liu, Lei Feng, and Hongxin Wei. Does confidence calibration improve conformal prediction? *Transactions on Machine Learning Research*, 2025. ISSN 2835-8856.
- Yu Xia, Jingru Fan, Weize Chen, Siyu Yan, Xin Cong, Zhong Zhang, Yaxi Lu, Yankai Lin, Zhiyuan Liu, and Maosong Sun. Agentrm: Enhancing agent generalization with reward modeling. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pp. 19277–19290, Vienna, Austria, 2025. Association for Computational Linguistics. doi: 10.18653/v1/2025.acl-long.945. URL <https://aclanthology.org/2025.acl-long.945/>.
- Liangru Xie, Hui Liu, Jingying Zeng, Xianfeng Tang, Yan Han, Chen Luo, Jing Huang, Zhen Li, Suhang Wang, and Qi He. A survey of calibration process for black-box llms. *arXiv preprint arXiv:2412.12767*, 2024.
- Miao Xiong, Zhiyuan Hu, Xinyang Lu, Yifei Li, Jie Fu, Junxian He, and Bryan Hooi. Can LLMs express their uncertainty? an empirical evaluation of confidence elicitation in LLMs. In *The Twelfth International Conference on Learning Representations*, 2024. URL <https://openreview.net/forum?id=gjeQKFxFpZ>. Poster.
- Tianyang Xu, Dan Zhang, Kushan Mitra, and Estevam Hruschka. Verification-aware planning for multi-agent systems. In *Proceedings of the 19th Conference of the European Chapter of the Association for Computational Linguistics (Volume 1: Long Papers)*, pp. 7528–7546, Rabat, Morocco, 2026. Association for Computational Linguistics. doi: 10.18653/v1/2026.eacl-long.353. URL <https://aclanthology.org/2026.eacl-long.353/>.
- Shi-Qi Yan, Jia-Chen Gu, Yun Zhu, and Zhen-Hua Ling. Corrective retrieval augmented generation. *arXiv preprint arXiv:2401.15884*, 2024. URL <https://arxiv.org/abs/2401.15884>.

-
- Xiao-Wen Yang, Xuan-Yi Zhu, Wen-Da Wei, Ding-Chu Zhang, Jie-Jing Shao, Zhi Zhou, Lan-Zhe Guo, and Yu-Feng Li. Step back to leap forward: Self-backtracking for boosting reasoning of language models. *arXiv preprint arXiv:2502.04404*, 2025. URL <https://arxiv.org/abs/2502.04404>.
- Shunyu Yao, Dian Yu, Jeffrey Zhao, Izhak Shafran, Thomas L Griffiths, Yuan Cao, and Karthik Narasimhan. Tree of thoughts: Deliberate problem solving with large language models. In *NeurIPS*, 2023.
- Zijun Yao, Weijian Qi, Liangming Pan, Shulin Cao, Linmei Hu, Liu Weichuan, Lei Hou, and Juanzi Li. Seakr: Self-aware knowledge retrieval for adaptive retrieval augmented generation. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pp. 27022–27043, 2025.
- Zhangyue Yin, Qiushi Sun, Qipeng Guo, Jiawen Wu, Xipeng Qiu, and Xuan-Jing Huang. Do large language models know what they don’t know? In *Findings of the Association for Computational Linguistics: ACL 2023*, pp. 8653–8665, Toronto, Canada, 2023. Association for Computational Linguistics. doi: 10.18653/v1/2023.findings-acl.551. URL <https://aclanthology.org/2023.findings-acl.551/>.
- Hee Suk Yoon, Eunseop Yoon, Mark A Hasegawa-Johnson, Sungwoong Kim, and Chang D Yoo. Confpo: Exploiting policy model confidence for critical token selection in preference optimization. In *Forty-second International Conference on Machine Learning*, 2025.
- Michael J. Zellinger and Matt Thomson. Rational tuning of LLM cascades via probabilistic modeling. *Transactions on Machine Learning Research*, 2025. ISSN 2835-8856.
- Yuanzhao Zhai, Han Zhang, Yu Lei, Yue Yu, Kele Xu, Dawei Feng, Bo Ding, and Huaimin Wang. Uncertainty-penalized reinforcement learning from human feedback with diverse reward lora ensembles. *arXiv preprint arXiv:2401.00243*, 2024. URL <https://arxiv.org/abs/2401.00243>.
- Dan Zhang, Sining Zhoubian, Ziniu Hu, Yisong Yue, Yuxiao Dong, and Jie Tang. ReST-MCTS*: LLM self-training via process reward guided tree search. *arXiv preprint arXiv:2406.03816*, 2024a. URL <https://arxiv.org/abs/2406.03816>.
- Hanning Zhang, Shizhe Diao, Yong Lin, Yi R Fung, Qing Lian, Xingyao Wang, Yangyi Chen, Heng Ji, and Tong Zhang. R-Tuning: Instructing large language models to say ‘i don’t know’. In *NAACL*, 2024b.
- Hongxiang Zhang, Hao Chen, Muhao Chen, and Tianyi Zhang. Active layer-contrastive decoding reduces hallucination in large language model generation. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, pp. 3028–3046, 2025a.
- Ling Zhang, Xianliang Yang, Juwon Yu, Park Cheonyoung, Lei Song, and Jiang Bian. Holdout-loss-based data selection for llm finetuning via in-context learning. *arXiv preprint arXiv:2510.14459*, 2025b.
- Tuo Zhang, Asal Mehradfar, Dimitrios Dimitriadis, and Salman Avestimehr. Leveraging uncertainty estimation for efficient llm routing. *arXiv preprint arXiv:2502.11021*, 2025c.
- Qinjian Zhao, Jiaqi Wang, Zhiqiang Gao, Zhihao Dou, Belal Abuhaija, and Kaizhu Huang. Safebehavior: Simulating human-like multistage reasoning to mitigate jailbreak attacks in large language models. *arXiv preprint arXiv:2509.26345*, 2025. doi: 10.48550/arXiv.2509.26345. URL <https://arxiv.org/abs/2509.26345>.
- Qihuang Zhong, Liang Ding, Li Shen, Juhua Liu, Bo Du, and Dacheng Tao. Revisiting knowledge distillation for autoregressive language models. *arXiv preprint arXiv:2402.11890*, 2024.
- Andy Zhou, Kai Yan, Michal Shlapentokh-Rothman, Haohan Wang, and Yu-Xiong Wang. Language agent tree search unifies reasoning, acting, and planning in language models. In *Proceedings of the 41st International Conference on Machine Learning*, volume 235 of *Proceedings of Machine Learning Research*, pp. 62138–62160. PMLR, 2024. URL <https://proceedings.mlr.press/v235/zhou24r.html>.

-
- Chunting Zhou, Pengfei Liu, Puxin Xu, Srinivasan Iyer, Jiao Sun, Yuning Mao, Xuezhe Ma, Avia Efrat, Ping Yu, Lili Yu, Susan Zhang, Gargi Ghosh, Mike Lewis, Luke Zettlemoyer, and Omer Levy. LIMA: Less is more for alignment. In *Advances in Neural Information Processing Systems 36 (NeurIPS 2023)*, 2023. URL https://proceedings.neurips.cc/paper_files/paper/2023/hash/ac662d74829e4407ce1d126477f4a03a-Abstract-Conference.html.
- Zhanke Zhou, Xiangyu Lu, Chentao Cao, Brando Miranda, Tongliang Liu, Bo Han, and Sanmi Koyejo. Codapo: Confidence and difficulty-adaptive policy optimization for post-training language models. In *2nd AI for Math Workshop@ ICML 2025*, 2025a.
- Ziang Zhou, Tianyuan Jin, Jieming Shi, and Qing Li. SteerConf: Steering LLMs for confidence elicitation. In *The Thirty-Ninth Annual Conference on Neural Information Processing Systems*, 2025b. URL <https://openreview.net/forum?id=5sgK63Zshg>. Poster.
- Yun Zhu, Jia-Chen Gu, Caitlin Sikora, Ho Ko, Yinxiao Liu, Chu-Cheng Lin, Lei Shu, Liangchen Luo, Lei Meng, Bang Liu, and Jindong Chen. Accelerating inference of retrieval-augmented generation via sparse context selection. *arXiv preprint arXiv:2405.16178*, 2024. URL <https://arxiv.org/abs/2405.16178>.
- Hanna Zubkova, Ji-Hoon Park, and Seong-Whan Lee. Sugar: Leveraging contextual confidence for smarter retrieval. In *ICASSP 2025-2025 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, pp. 1–5. IEEE, 2025.

A Related Surveys on Confidence, Calibration, and Hallucination

Several prior surveys examine confidence, uncertainty, calibration, hallucination, or abstention in LLMs. Their perspectives are complementary to ours, but their primary question is usually how a signal should be estimated, calibrated, diagnosed, or interpreted. By contrast, this survey asks how confidence-related signals are *used* to govern system behavior. Table 7 summarizes that distinction at a high level.

Survey	Primary Focus	Confidence Perspective	Coverage	Relation to Ours
Geng et al. (2024)	Confidence estimation and calibration	Confidence as an object to estimate and calibrate	LLM confidence methods and evaluation	Estimator-centric; does not organize downstream control decisions
Xie et al. (2024)	Black-box calibration	Confidence elicitation, self-reflection, and recalibration without logits	Black-box LLM confidence calibration	Calibration-centric; narrower than lifecycle-wide control
Liu et al. (2025c)	Uncertainty quantification and calibration	Uncertainty decomposed into multiple sources	Multi-stage LLM UQ and evaluation	UQ-centric; complements our control view
Shorinwa et al. (2025)	Uncertainty quantification	Broad taxonomy of uncertainty sources and diagnostics	LLM uncertainty methods and tasks	UQ-centric; emphasizes estimation rather than action
Rawte et al. (2023)	Hallucination in foundation models	Detection and mitigation signals organized by failure type	Text, image, video, and audio foundation models	Failure-mode survey, not confidence-guided control taxonomy
Huang et al. (2025c)	Hallucination in LLMs	Hallucination causes, detection, and mitigation	LLM lifecycle from pre-training to inference	Organized around one failure mode rather than one control signal
Tonmoy et al. (2024)	Hallucination mitigation	Intervention methods for reducing hallucination	LLM mitigation strategies, including RAG and decoding	Mitigation-centric; narrower than our control framing
Wen et al. (2025)	Abstention and “knowing what you don’t know”	Confidence as a basis for refusal or deference	Pre-training, fine-tuning, alignment, and inference	Closest to ours, but centered on one decision family
This survey	Confidence utilization	Confidence as a control signal	Training, inference, routing, RAG, risk, and agentic control	Cross-domain framework for confidence-guided decisions

Table 7: Related surveys on confidence, uncertainty, hallucination, and abstention in LLMs. Prior surveys mainly organize the literature around estimation, calibration, failure analysis, or a single decision family. This survey instead treats confidence-related signals as control inputs that govern downstream system behavior across the LLM lifecycle.

The most direct predecessors are the confidence-estimation and uncertainty-quantification surveys. Geng et al. (2024) review confidence estimation and calibration for LLMs, while Xie et al. (2024) focus specifically on black-box calibration settings where internal logits may be unavailable. Liu et al. (2025c) and Shorinwa et al. (2025) broaden the discussion to uncertainty quantification, distinguishing different uncertainty sources and surveying calibration diagnostics, conformal tools, and related evaluation methods. These surveys provide essential background on how confidence-like signals are elicited, estimated, or calibrated, but they do not systematically trace how such signals are consumed by downstream policies once they are available. In the language of Section 2, they focus primarily on signal formation, whereas our emphasis is on the actions those signals support.

A second line of related work organizes the literature around hallucination rather than confidence. Rawte et al. (2023) survey hallucination across foundation models and modalities; Huang et al. (2025c) provide an LLM-specific view of hallucination causes, benchmarks, detection, and mitigation across the lifecycle; and Tonmoy et al. (2024) emphasize mitigation strategies such as retrieval, decoding constraints, and fine-tuning interventions. These surveys are highly relevant to our discussions of groundedness, failure detection, and abstention, especially in Section 6 and Section 7. However, they are organized around a particular failure

mode. Our survey instead treats hallucination detection as one instance of a broader pattern in which confidence-related signals trigger decisions such as filtering, retrieval, abstention, escalation, or revision.

Wen et al. (2025) are closest in spirit to our risk-management discussion because they treat abstention as a first-class action and analyze how models can decide when not to answer. Their contribution is important precisely because it shifts attention from confidence estimation alone to a downstream decision. Our framing extends that idea in two ways. First, we treat abstention as one member of a broader family of confidence-guided actions that also includes selection, routing, retrieval control, search control, and aggregation. Second, we follow those actions across six parallel domains of the LLM lifecycle rather than centering a single decision family.

Summary. Prior surveys provide strong foundations on confidence estimation, uncertainty quantification, hallucination analysis, and abstention. Our contribution is to connect these strands through a unified control perspective: confidence-related signals are not only quantities to be estimated, but instruments for governing behavior across the full LLM lifecycle.